STATICALCS: AN INTERACTIVE WEB-BASED LEARNING TOOL WITH INTEGRATED CALCULATORS FOR STATICS OF RIGID BODIES

An Undergraduate Thesis
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Bachelor of Science in Civil Engineering

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Contents

List of Figures

List of Tables

Chapter 1
Introduction

1.1 Background of the Study

Engineering Mechanics is one of the fundamental components of engineering education, and Statics of Rigid Bodies serves as its essential foundation. The course develops the analytical skills needed to understand how forces and moments act on bodies at rest, making it a prerequisite for more advanced engineering subjects. In the Civil Engineering curriculum of Mindanao State University – General Santos (MSU–Gensan), *Statics of Rigid Bodies (ENS161)* is a core subject required for academic progression in the engineering program.

Despite its importance, many students find Statics difficult due to its mathematical requirements and abstract nature. Students often struggle with force systems, moments, and the construction of accurate *Free Body Diagrams (FBDs)*, which are necessary for solving Statics of Rigid Bodies problems. I. Salami and Perry found that learners commonly face conceptual and procedural difficulties in Statics, including challenges in visualizing forces and connecting equations to physical situations. These difficulties are further intensified by weak mathematical foundations, particularly in algebra and trigonometry, which are essential for problem-solving in Statics. Felix noted that incoming college students often exhibit declining mathematical readiness, affecting their ability to perform well in computation-heavy subjects.

Additionally, affective factors such as math anxiety contribute to students' performance challenges. Incierto et al. found that higher levels of math anxiety are associated with lower academic achievement, indicating that many students struggle not only with understanding concepts but also with the emotional stress associated with solving mathematical problems. Research also shows that appropriate tools, such as calculators and structured digital aids, can reduce computational errors and alleviate anxiety by allowing learners to focus on conceptual understanding (Segarra and Cabrera-Martínez).

Traditional lecture-based instruction, which relies heavily on verbal explanation and boardwork, often provides limited visualization and interactivity. Modern learners, particularly those belonging to Generation Z, respond more positively to technology-supported learning environments. Szymkowiak et al. emphasized that digital-native students benefit from online tools, multimedia resources, and interactive platforms that enhance engagement and comprehension.

Given these challenges, students benefit from supplementary learning tools that reinforce classroom instruction, enhance visualization, and support step-by-step problem solving. In response, the study developed *StatiCalcs*, a web-based interactive learning tool designed to help MSU–Gensan Civil Engineering students understand Statics of Rigid Bodies. The platform provides topic-aligned resources and integrated calculators that generate step-by-step solutions and visual representations, offering an accessible and technology-supported supplement to traditional instruction.

1.2 Statement of the Problem

This study aims to develop a web-based learning tool, *StatiCalcs*, specifically designed for the subject Statics of Rigid Bodies. After the development of the application, the study seeks to determine the level of perception of Civil Engineering students and instructors of MSU–General Santos regarding its usability, accessibility, and user satisfaction.

Specifically, this study aims to answer the following questions:

1. What is the level of perception of Bachelor of Science in Civil Engineering students of MSU–General Santos regarding *StatiCalcs* in terms of:

    1.1 Usability;

1.2 Accessibility; and

1.3 Satisfaction?

2. What is the level of perception of Bachelor of Science in Civil Engineering instructors of MSU–General Santos regarding *StatiCalcs* in terms of:

   2.1 Usability;

   2.2 Accessibility; and

   2.3 Satisfaction?

3. Is there a significant difference between the level of perception of Bachelor of Science in Civil Engineering students and instructors regarding *StatiCalcs* in terms of usability, accessibility, and satisfaction?

4. What is the level of preference of Bachelor of Science in Civil Engineering students and instructors for different learning resources in studying Statics of Rigid Bodies?

Null Hypothesis

Ho: There is no significant difference between the level of perception of Bachelor of Science in Civil Engineering students and instructors regarding *StatiCalcs* in terms of usability, accessibility, and satisfaction.

1.3 Scope and Limitations

This study will focus on the design, development, and evaluation of a web-based learning tool, *StatiCalcs*, intended to assist Bachelor of Science in Civil Engineering students of Mindanao State University – General Santos in studying the subject Statics of Rigid Bodies. The website will cover all major topics in Statics, including force systems, equilibrium, structures, centroids, moments of inertia, and friction. It will integrate step-by-step solutions and calculators to help users understand and solve problems interactively. The evaluation of the tool will focus on three aspects: usability, accessibility, and user satisfaction.

The respondents of the study will be limited to Bachelor of Science in Civil Engineering students, specifically second-year students currently enrolled in Statics of Rigid Bodies and students who have already taken the subject, as well as instructors teaching the same course in the College of Engineering of MSU–General Santos during the Academic Year 2025–2026. Data will be collected through survey questionnaires administered after respondents have explored and used the developed website.

The study will not measure students' academic performance or actual improvement in examination results. It will also exclude students and instructors from other engineering programs. Moreover, the study will focus solely on perceptions of the usability and effectiveness of the developed tool. It will not account for external factors such as internet connectivity, learning motivation, or respondents' prior knowledge.

The scope of this project will include the development of a website using modern web technologies, focusing on both frontend and backend functionality. The backend of the website will be built in Java, handling application logic and processing user interactions. The frontend interface will be designed with Tailwind CSS, ensuring a responsive, modern, and visually consistent user experience. Version control and collaboration will be managed through GitHub, while deployment and hosting will be automated via Netlify. ChatGPT will be utilized as a supplementary tool to assist in web design decisions, code arrangement, and debugging, providing guidance and improving workflow efficiency.

Scope of the Study

This study focuses on the design, development, and evaluation of a web-based learning tool, *StatiCalcs*, intended to assist Bachelor of Science in Civil Engineering students of

Mindanao State University – General Santos in studying the subject Statics of Rigid Bodies.

Coverage of the Study

The website will cover major topics in Statics, including force systems, equilibrium, structures, centroids, and moments of inertia. It will integrate step-by-step solutions and calculators, and downloadable PDF files to help users understand and solve problems interactively. The evaluation of the tool will focus on three aspects: usability, accessibility, and user satisfaction.

Participants

The respondents of the study will be limited to Bachelor of Science in Civil Engineering students, specifically second-year students currently enrolled in Statics of Rigid Bodies and students who have already taken the subject, as well as instructors teaching the same course in the College of Engineering of MSU–General Santos during the Academic Year 2025–2026. Data will be collected through survey questionnaires administered after respondents have explored and used the developed website.

Time and Setting

The study will be conducted during the Academic Year 2025–2026 within the College of Engineering of Mindanao State University – General Santos.

Website Features Covered

The scope of this project will include the development of a website using modern web technologies, focusing on both frontend and backend functionality. The backend of the website will be built in Java, handling application logic and processing user interactions. The frontend interface will be designed with Tailwind CSS, ensuring a responsive, modern, and visually consistent user experience. Version control and collaboration will be managed through GitHub, while deployment and hosting will be automated via Netlify. ChatGPT will be utilized as a supplementary tool to assist in web design decisions, code arrangement, and debugging, providing guidance and improving workflow efficiency.

Limitations of the Study

Technical Limitations

The backend will be limited to Java, which may restrict integration with other programming languages or frameworks without additional configuration. The frontend design will depend on Tailwind CSS, and deviations from its utility-first approach may require additional custom CSS. While GitHub and Netlify will streamline version control and deployment, they will require internet connectivity and account setup, which could pose challenges in offline or restricted environments. Furthermore, because the website is hosted on a website hosting platform and not on a dedicated server-based environment, it cannot store or manage user data on the web. Additionally, the website's functionality and performance are constrained by the features provided by the chosen technologies and hosting platform, limiting scalability for highly complex or resource-intensive applications.

Educational Limitations

The study will not measure students' academic performance or actual improvement in examination results. It will also exclude students and instructors from other engineering programs. The study will focus solely on perceptions of the usability and effectiveness of the developed tool and will not account for external factors such as learning motivation or respondents' prior knowledge. Furthermore, the website is designed to supplement learning and not to replace foundational instruction, as users are expected to have a basic background and understanding of each major topic in Statics of Rigid Bodies to fully utilize the features of the website.

Research Limitations

The study will not account for external factors such as internet connectivity, learning motivation, or respondents' prior knowledge that may influence respondents' perceptions and evaluations of the tool. The findings will be based solely on data collected through survey questionnaires administered after respondents have explored and used the developed website.

System Limitations

The problems covered in the website are limited to basic problems within each major topic in Statics of Rigid Bodies and will not include special cases or advanced problem variations. Additionally, the calculators are designed to process inputs based on an already analyzed problem, meaning users must first thoroughly analyze the problem before inputting values into the calculator. The website does not automate problem analysis or diagram generation, and therefore requires the user to apply their own understanding and judgment prior to using the tools. ChatGPT will serve only as a supportive tool and does not replace the developer's judgment or expertise. The website's overall functionality and performance are constrained by the features and capabilities provided by the chosen technologies and hosting platform, limiting scalability for highly complex or resource-intensive applications.

1.4 Significance of the Study

This study will focus on developing a web-based learning tool, *StatiCalcs*, to support the teaching and learning of Statics of Rigid Bodies among Bachelor of Science in Civil

Engineering students at MSU–General Santos. The results and outcomes of this research may benefit the following:

Students. The developed website may serve as a supplementary learning tool, allowing students to practice problem-solving in Statics of Rigid Bodies through an interactive, user-friendly platform. As learners today are inclined toward technology-based education, *StatiCalcs* may provide an accessible and engaging way to enhance understanding and reinforce classroom instruction.

Teachers. The tool may be integrated into teaching strategies to improve the delivery of complex statics concepts. It may assist instructors in providing visual and computational demonstrations that complement lectures and classroom exercises.

School Administration. The study may provide the institution with an example of how to develop and implement digital learning tools to improve academic performance. It may also encourage future initiatives that promote technology-driven learning in other engineering courses.

Future Researchers. This study may serve as a reference for future studies aiming to develop similar educational tools or further improve *StatiCalcs* by adding new features, expanding its scope, or evaluating its long-term impact on student learning.

1.5 Theoretical Framework

This study draws on three foundational frameworks to guide the evaluation of digital learning environments. The first is Nielsen's Usability Heuristics, a widely recognized set of principles that describe the characteristics of an effective, user-friendly interface. These heuristics highlight the importance of intuitive navigation, clear information presentation, consistency, error prevention, and efficiency of use. By applying these principles, the study evaluates how easily users can interact with StatiCalcs and how effectively the tool supports learning tasks in Statics. This theoretical support strengthens the assessment of usability by linking user experience to established standards in interface and system design.

The second framework is the Web Content Accessibility Guidelines (WCAG 2.1), which provides internationally accepted criteria for ensuring that digital platforms are accessible to all users regardless of device limitations, environmental conditions, or individual differences. WCAG emphasizes perceivability, operability, understandability, and robustness—principles that ensure that content is readable, navigable, and functional across various contexts. Alongside this, the User Satisfaction Model explains that satisfaction arises when users perceive a system as useful, reliable, and aligned with their expectations. In this study, WCAG supports the evaluation of accessibility, while the User Satisfaction Model supports the measurement of satisfaction with the tool as a supplementary learning resource. Together, these theories establish a strong foundation for examining StatiCalcs in terms of usability, accessibility, and user satisfaction.

1.6 Conceptual Framework

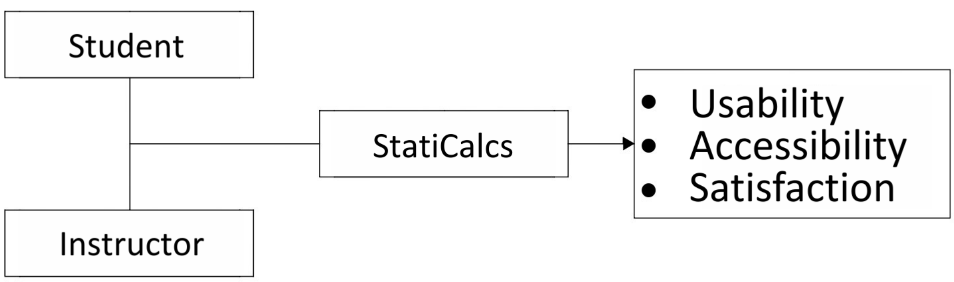

Figure 1.1: Conceptual Framework

This study is anchored in a conceptual framework that describes the relationship between users of the system and the web-based learning tool *StatiCalcs*, and how their interaction with the platform leads to the evaluation of three key system attributes: usability, accessibility, and satisfaction. The independent variables are the students and instructors, while the dependent variables are their level of perception in terms of usability, accessibility, and satisfaction.

StatiCalcs functions as the central intervention, offering computational features and step-by-step problem-solving support. The quality of users' interactions with the system is evaluated through three outcome variables: usability, accessibility, and satisfaction. Usability pertains to the efficiency and ease with which users navigate and perform tasks within the platform. Accessibility evaluates the system's ability to accommodate diverse users, devices, and learning conditions. Satisfaction measures the overall acceptance of the tool and the extent to which it meets user expectations.

Overall, the framework posits that students' and teachers' experiences with *StatiCalcs* directly shape their perceptions of its usability, accessibility, and satisfaction, thereby informing the system's effectiveness as a supplementary learning tool.

1.7 Definition of Terms

Accessibility — The extent to which Bachelor of Science in Civil Engineering students enrolled in Statics of Rigid Bodies and their instructors can easily access, navigate, and use the developed website, *StatiCalcs*, regardless of ability, to support effective learning and teaching.

Bachelor of Science in Civil Engineering Students	Refers to the respondents of the study who are currently enrolled in or have previously completed the course Statics of Rigid Bodies in the Academic Year 2025–2026. These participants will provide data regarding their perception of the usability, accessibility, and satisfaction of the developed website.
Bachelor of Science in Civil Engineering Instructors	Refers to the respondents of the study who are teaching the course Statics of Rigid Bodies during the Academic Year 2025–2026. Their feedback will contribute to evaluating the effectiveness, usability, and accessibility of the developed website as a supplementary learning tool.
Digital Natives	Individuals who grew up in the age of digital technology and are comfortable using computers, the Internet, and web-based applications. They represent the target population of this study.
Dynamics of Rigid Bodies	A branch of mechanics that examines the motion of non-deformable bodies under applied forces, including translational and rotational motion, using Newton's laws and principles of energy and momentum. Prerequisite: Statics of Rigid Bodies.

Free Body Diagram (FBD)	A Free Body Diagram (FBD) is a simplified graphical representation of a body or system isolated from its surroundings, showing all external forces and moments acting on it. It is used to visualize and solve problems involving equilibrium.
Mechanics of Deformable Bodies	Also known as the mechanics of materials, this field studies the behavior of solid bodies that deform under applied forces. It analyzes how stresses and strains are distributed within materials and relates these deformations to material properties. Prerequisite: Statics of Rigid Bodies.
Perception	The way students and instructors interpret and evaluate their experiences with a learning tool, concept, or environment. In this study, it refers to how they perceive the usability, accessibility, and effectiveness of the developed website.
Statics of Rigid Bodies	A branch of engineering mechanics that studies forces and their effects on rigid bodies, focusing on conditions of equilibrium where the sum of all forces and moments equals zero. This course is the main focus of the study.

Satisfaction The degree to which Bachelor of Science in Civil Engineering students enrolled in Statics of Rigid Bodies and their instructors feel content or fulfilled with the developed website, *StatiCalcs*, reflecting how well it meets their learning needs and expectations.

Usability The degree to which Bachelor of Science in Civil Engineering students enrolled in Statics of Rigid Bodies and their instructors can effectively, efficiently, and satisfactorily use the developed website, *StatiCalcs*, to achieve their learning and teaching goals.

Chapter 2
Review of Related Literature

The Review of Related Literature discusses previous studies and theories relevant to this research. It focuses on the challenges students face in learning Statics of Rigid Bodies, the use of web-based learning tools, and learners' perceptions of technology-assisted instruction. This review highlights the need for an interactive platform like *StatiCalcs* to support effective learning and teaching.

2.1 Statics of Rigid Bodies in Engineering Education

Statics of Rigid Bodies is a fundamental area of Engineering Mechanics concerned with understanding how forces and moments act on bodies that remain in equilibrium. As a foundational subject, Statics develops the analytical reasoning necessary for advanced topics such as Dynamics, Strength of Materials (Mechanics), Structural Analysis, and Fluid Mechanics.

In the Civil Engineering curriculum of Mindanao State University – General Santos (MSU-Gensan), as stated in MSU Board Resolution No. 375, s. 2017, Statics of Rigid Bodies (ENS161) is formally designated as a foundation course and a prerequisite for higher engineering subjects. Its role as a gateway subject underscores the need for students to develop strong problem-solving, visualization, and mathematical reasoning skills before moving on to more complex engineering analyses.

For Civil Engineering students, mastery of Statics is essential because it directly relates to designing structures that safely resist loads. Failure to grasp the principles of force systems, equilibrium, and moment calculations may hinder students' academic progression and reduce their preparedness for professional engineering practice (I. E. Salami et al.). Its foundational role in engineering education highlights the need for effective teaching approaches and supportive learning resources.

2.2 Conceptual Difficulties in Learning Statics

Learning Statics poses substantial conceptual challenges for students, mainly because the subject requires interpreting forces and interactions that are not directly observable. Students frequently struggle to visualize force directions, understand moment effects, and connect theoretical principles to real-world mechanical behavior. These conceptual gaps often lead to incorrect assumptions about how forces act on objects and how equilibrium conditions are achieved.

One of the most common sources of difficulty is the abstraction involved in translating a real object into a simplified engineering model. Students must isolate bodies, idealize supports, and mentally detach systems from their environment, steps essential to constructing valid Free Body Diagrams (FBDs). When learners are unable to visualize these relationships clearly, their problem-solving accuracy decreases significantly.

Misconceptions about force components, resultant forces, and moment calculations further contribute to students' struggles. Many learners also find it challenging to distinguish between couples, force systems, and distributed loads. These conceptual difficulties suggest the need for instructional approaches that enhance visualization, provide immediate feedback, and reduce students' dependence on rote memorization.

2.3 Procedural and Analytical Challenges

In addition to conceptual issues, I. Salami and Perry emphasized that students often face procedural difficulties when applying Statics principles to computational tasks. Many struggle to correctly break forces into components, apply equilibrium equations, or follow systematic steps when solving problems. These challenges frequently arise from gaps in prior mathematical knowledge, particularly in algebra and trigonometry.

Procedural errors also occur when students attempt to construct FBDs without fully understanding the physical systems involved. Incorrect or incomplete diagrams often lead to errors that persist through the entire solution process. According to I. Salami and Perry, procedural weaknesses combined with conceptual misunderstandings contribute to persistent difficulties in Statics, even among higher-year students.

Students also tend to rely on memorized formulas without understanding their derivations or applications. This approach hampers their ability to transfer knowledge to unfamiliar problem types or adapt to more complex engineering tasks. Such procedural weaknesses highlight the need for supplementary learning resources that guide students step by step and reinforce systematic problem-solving habits.

2.4 Mathematical Readiness in Statics

A strong mathematical foundation is a prerequisite for success in Statics of Rigid Bodies. However, many incoming engineering students lack proficiency in algebra, geometry, trigonometry, and vector operations, skills essential for analyzing forces and moments. Felix reports that declining math preparedness among first-year students negatively affects performance in computation-heavy engineering subjects.

Wenceslao conducted a study assessing the mathematical and analytical readiness of incoming first-year engineering students in Eastern Visayas, Philippines. The study revealed that 57% of students were not mathematically college-ready, demonstrating significant deficiencies in key areas such as calculus, trigonometry, and algebra, which are essential for problem-solving in engineering disciplines like Statics. Students struggled particularly with integration, derivatives, and trigonometric identities, indicating gaps in foundational skills needed to manipulate equations, solve simultaneous equations, resolve forces, and understand vector relationships. These weaknesses often lead to frustration, disengagement, and errors in calculating resultants, determining moment arms, and resolving angled forces.

The findings underscore the need for early interventions, remedial support, digital or supplementary learning tools that scaffold mathematical procedures, reducing the cognitive burden of complex computations and allowing learners to focus on conceptual understanding and practical application in Statics. Digital resources that automate or guide mathematical procedures may help students focus more on conceptual interpretation and problem comprehension.

2.5 Math Anxiety and Its Impact on Engineering Students

Math anxiety is a widely documented factor that affects students' performance in technical subjects. Incierto et al. found that students with higher levels of anxiety tend to perform poorly in mathematics-related courses, including engineering subjects that rely heavily on quantitative skills. This emotional barrier often causes learners to avoid practicing mathematical problems, ultimately reducing their mastery of essential skills.

In Statics, where problem-solving requires applying multiple mathematical steps, anxiety can impair students' ability to think clearly, interpret diagrams, and follow procedural logic. Reducing math anxiety is therefore an important aspect of improving student performance.

Several studies suggest that allowing students to use calculators, digital tools, or guided computational aids can mitigate anxiety by reducing the pressure associated with manual calculations. Segarra and Cabrera-Martínez found that structured calculator use improves students' confidence and accuracy, enabling them to concentrate more on conceptual understanding than on arithmetic manipulation.

2.6 Visualization and Spatial Ability in Engineering

Spatial reasoning is a key competency in engineering, particularly in Statics, where students must interpret diagrams, imagine 3D force interactions, and understand geometric relationships. Learners with weak spatial ability often struggle with vector representation, force decomposition, and moment visualization (Fontaine and Vallabh).

Visualization tools, including interactive diagrams and simulations, help develop these skills by allowing students to manipulate variables and observe changes in real time. Zeichner argues that simulations enhance student motivation and conceptual understanding, particularly in visually demanding subjects such as Statics. Given that Statics requires a combination of spatial and analytical skills, web-based learning tools that incorporate visualization are essential for supporting diverse learning needs.

De La Hoz et al. emphasized the effectiveness of visualization-based platforms in Statics learning through the integration of the spatial simulation tool Hapstatics. In their study, students interacted with worked examples while self-explaining each solution step, which improved conceptual understanding of static equilibrium. The researchers concluded that the integration of visualization tools is essential for students to achieve a deeper understanding of the physical behaviors illustrated in Statics exercises, as these environments help learners identify forces, reactions, and structural behavior more accurately.

They further noted that self-explanation practices significantly contribute to conceptual change, as students who verbalize and justify their solution steps demonstrate better problem-solving performance and error recognition. While this process may require prompting and structured guidance, interactive digital environments support these outcomes by encouraging explanation, reflection, and correction of misconceptions. This approach aligns with the purpose of *StatiCalcs*, which similarly provides visual and guided analytical support to strengthen conceptual understanding beyond text-based explanation.

2.7 Importance of Free Body Diagram Construction

The Free Body Diagram (FBD) is one of the most critical tools in Engineering Mechanics. It serves as the foundation for analyzing force interactions, solving equilibrium equations, and understanding mechanical behavior. Despite its importance, many students struggle to construct accurate FBDs, often omitting forces, misrepresenting directions, or misidentifying support reactions (I. Salami and Perry).

Errors in FBD construction typically result in incorrect equilibrium equations, making the entire solution process invalid. Researchers emphasize the need for learning tools that support FBD visualization, provide real-time diagrams, and reinforce correct diagramming techniques. Visual learning tools and graphical simulations have been shown to improve performance and deepen understanding in Statics.

2.8 Technology-Based Learning for Engineering Student

The concept of digital natives, introduced by Prensky, describes individuals who have grown up immersed in digital technology, such as computers, smartphones, and the internet. This group generally includes those born from the 1980s onward, commonly identified as Millennials, Generation Z, and Generation Alpha. These generations grew up during the rapid advancement of technology, which has profoundly influenced their lifestyles, behaviors, and learning preferences. Since they have never known a world without digital connectivity, technology has become an integral part of their identity and daily lives. According to Szymkowiak et al., educators should combine traditional teaching methods with modern, internet-based learning tools, including mobile applications and online videos, to accommodate the learning preferences. Zeichner similarly found that learning through simulation fosters a deeper understanding of abstract principles and concepts compared with conventional instruction. These findings are supported by

De La Hoz et al., who observed that traditional lecture-based teaching in statics often fails to address students' learning barriers and problem-solving difficulties.

Also, a comprehensive literature review by Tawafak et al. highlights that the continuous exposure of Gen Z to digital environments has transformed their learning preferences and expectations in higher education. Their familiarity with interactive, technology-driven systems opens opportunities for educators to design more engaging, adaptive digital learning platforms. Collectively, these studies emphasize the importance of integrating interactive, technology-enhanced approaches into higher education to meet the learning needs of digital natives in modern academic settings and to improve student engagement, comprehension, and analytical skills, especially in conceptually demanding courses such as Statics of Rigid Bodies.

Technology-based learning enhances engagement, improves retention, and provides flexibility for self-paced study. Web-based learning tools have been shown to promote active learning, encourage repeated practice, and support independent problem-solving. These tools can supplement traditional lectures, which often lack interactivity and visualization. Given these advantages, integrating technology into Statics instruction is essential to address students' difficulties and enhance their learning experience.

2.9 Digital Learning Tools and Student Engagement

Digital learning tools have become increasingly prominent in education due to their ability to enhance motivation, interaction, and learner engagement. These tools provide flexible access to learning resources, enabling students to study at their own pace and revisit instructional content as needed. Pedraza and Canoy emphasize that mobile learning applications significantly improve student engagement by offering features such as instant feedback, interactive activities, and convenient access to lessons even outside classroom hours. Their findings show that such tools sustain learners' interest and encourage continuous participation, especially when digital platforms are easy to use and readily accessible.

Similarly, Hassan et al. highlight that digital learning tools are most effective when designed to be simple, visually clear, and easy to navigate. Their study on usability for slow learners demonstrates that clear interfaces, minimal complexity, and structured layouts help reduce cognitive load, allowing students to focus on learning tasks rather than struggling with the platform. These insights suggest that usability and accessibility directly influence engagement. Students are more likely to remain motivated when digital tools support smooth interaction and reduce barriers to understanding.

Collectively, these studies show that digital learning tools foster engagement when they are accessible, user-friendly, and supportive of students' learning processes. The findings from Pedraza and Canoy, and from Hassan et al., consistently highlight the importance of usability, accessibility, and a positive user experience in promoting active and continuous participation in digital learning environments. These insights provide strong justification for evaluating web-based educational tools, such as StatiCalcs, on their usability, accessibility, and ability to sustain student engagement.

2.10 Web-Based Learning in Engineering Education

Web-based learning has become a significant trend in modern education due to its accessibility, flexibility, and multimedia integration capabilities. Through internet-based platforms, students can access materials anytime, practice problems repeatedly, and receive immediate feedback. This is especially useful in technical subjects that require repeated exposure to problem-solving procedures.

Research by Abumandour on the potential of e-learning as an educational system for engineering topics highlighted the rapid technological advancements of the 21st century, which have transformed educational delivery methods. The researcher noted that advances in technology have led to the emergence of e-learning as a modern instructional approach widely adopted by educational institutions, public organizations, and academic libraries. The researcher highlighted that engineering education is increasingly moving toward a blended learning model that effectively integrates traditional "face-to-face" instruction with computer-assisted methodologies and internet-based learning. This hybrid approach enhances accessibility, flexibility, and learner engagement.

The study also outlined several challenges and obstacles that stakeholders, including teachers, professors, and librarians, must address to develop and sustain effective e-learning systems fully. These challenges include ensuring technological readiness, maintaining instructional quality, and supporting both instructors and students in adapting to digital learning environments. The researcher further proposed recommendations to strengthen the connection between e-learning and engineering education, thereby promoting a more adaptive, innovative, and inclusive academic experience. The researcher highlights the growing need for digital tools in engineering education as institutions shift toward blended and online learning systems.

In the study of Söllradl, which aims to enhance digital learning in Engineering Mechanics across several departments, the researcher developed an automated, scalable system to generate individualized assignments for students, supporting a more efficient, interactive approach to learning. To accomplish this, the researcher used a Python-based framework that automatically generates, distributes, and solves assignments for beam structures. Instead of relying solely on numerical solutions, the system emphasized symbolic computation using the *Euler–Bernoulli beam theory* to formulate and solve linear systems of equations. This method provided learners with a clearer understanding of the mechanical behavior of beam structures by reinforcing analytical and theoretical principles rather than focusing only on numerical outputs. Through this approach, this study demonstrated how digitalization and automation can be effectively integrated into engineering education to promote personalized learning, reduce instructors' manual workload, and enhance students' analytical and problem-solving skills.

Another significant advancement in web-based learning was introduced by Gfrerer et al., who developed an automated correction system for individualized exercise assignments in Engineering Mechanics courses. Their work addresses long-standing challenges in large classes, where manually creating and grading assignments limits the variety of problems students can encounter. With smaller problem pools, learners may replicate solutions without developing genuine conceptual understanding, and instructors may struggle to assess independent problem-solving skills.

To overcome these issues, the researchers developed a scalable digital framework capable of generating, distributing, and automatically correcting personalized exercises in Statics, Strength of Materials, Dynamics, and Hydrostatics. Each student receives a unique set of problems, promoting academic integrity and encouraging active, self-directed learning. A quantitative survey among Statics of Rigid Bodies students showed strong acceptance of the tool, with many reporting enhanced reflective learning and improved engagement. The authors concluded that automated correction systems significantly improve efficiency in mechanics education by reducing instructor workload while supporting deeper learner understanding.

Web-based tools allow for unified access across devices, making them ideal supplements for engineering subjects that require visual illustrations and computational support. These features align perfectly with the needs of Statics students who benefit from guided examples, diagrams, and automated computations.

2.11 Related Systems and Existing Educational Tools

Several digital tools and online platforms have been developed to support learning in mathematics and engineering, but each has limitations when applied explicitly to Statics of Rigid Bodies. Symbolab, for instance, provides automated, step-by-step solutions for algebra, calculus, and trigonometric problems. A study by Paulin and Jean Baptiste aimed to determine the effectiveness of the Symbolab Calculator in improving second-year students' ability to solve trigonometric equations. The findings suggest that Symbolab can enhance students' problem-solving skills and understanding of mathematical concepts. However, despite these benefits, Symbolab lacks dedicated features for statics topics such as equilibrium, force systems, and truss structures, limiting its direct applicability in engineering Statics courses.

GeoGebra offers dynamic and interactive visualizations for geometry and algebra. Its visual approach enhances students' spatial reasoning and understanding of mathematical relationships. However, GeoGebra does not include built-in tools for engineering Statics, such as truss analysis or 2D/3D equilibrium calculations.

More specialized platforms, such as SkyCiv, provide advanced structural analysis tools for 2D and 3D frames and trusses. While highly useful for engineering applications, many features require paid subscriptions, reducing accessibility for undergraduate students.

Other digital innovations, such as the automated correction systems developed by Gfrerer et al., focus on generating individualized problem sets and automated grading in Engineering Mechanics courses. These systems enhance self-directed learning but primarily support assessment rather than conceptual understanding. Similarly, symbolic computation tools developed by Söllradl emphasize automated and symbolic solutions for beam structures in Strength of Materials, but do not provide comprehensive coverage of fundamental statics topics.

Although these tools contribute significantly to engineering education, none provide an integrated, free, and Statics-specific learning environment that combines step-by-step solutions, real-time computations, and dynamic visualizations. This gap underscores the need for a specialized platform like *StatiCalcs*, designed for undergraduate Statics students, providing an accessible, interactive, and comprehensive learning tool that addresses both conceptual understanding and practical problem-solving.

2.12 Summary of Literature

The literature consistently shows that Statics is conceptually demanding, procedurally complex, and dependent on strong mathematical foundations. Students struggle with visualization, FBD construction, force interactions, and equilibrium analysis. Traditional instruction alone often fails to address these learning gaps.

Research also emphasizes the importance of digital tools, visual learning, and web-based platforms in enhancing student understanding, especially for digital-native learners. Existing tools such as Symbolab, GeoGebra, and SkyCiv offer partial support but do not provide a comprehensive Statics learning environment.

The reviewed literature clearly indicates the need for an accessible, interactive, and specialized web-based tool to support students' learning in Statics. This provides strong justification for developing *StatiCalcs*, which integrates visualization, computation, and conceptual understanding to address the gaps identified in previous studies.

Chapter 3
Methodology

This chapter presented the research method, respondents, locale, instrument, and procedure used in the study. It explained how the research was conducted to determine the perceptions of Bachelor of Science in Civil Engineering students and instructors toward *StatiCalcs*. This interactive web-based learning tool was developed for the Statics of Rigid Bodies course.

3.1 Research Method

This study used a quantitative research method, which focused on collecting and analyzing numerical data to describe patterns, relationships, and differences among variables. According to Apuke, quantitative research involved a systematic investigation that used statistical techniques to produce objective and measurable results. This method was appropriate for the present study because it sought to gather data from Civil Engineering students and instructors to assess and compare their perceptions of the developed web-based learning tool, *StatiCalcs*.

Specifically, a correlational quantitative approach was applied. As defined by Creswell, this approach examined the relationship or difference between two or more variables to determine whether a significant association existed. In this study, it was used to determine the significant difference between the perceptions of students and instructors regarding the usability, accessibility, and satisfaction of *StatiCalcs*. This method allowed the researchers to statistically analyze the degree to which these two groups differed in their perceptions, providing insights into the effectiveness and acceptance of the developed learning tool across users.

3.2 Research Respondents

The respondents of this study consisted of Bachelor of Science in Civil Engineering students and civil engineering faculty members from the College of Engineering at Mindanao State University – General Santos.

The student respondents were selected from second-year to fourth-year civil engineering students who were either currently taking or had already completed the course *Statics of Rigid Bodies (ENS161).* These students were chosen to ensure that all participants had sufficient background knowledge of the subject, enabling them to provide relevant and informed feedback on the web-based learning tool, *StatiCalcs*.

The total number of eligible students was 449, composed of 129 second-year, 102 third-year, and 218 fourth-year students. The required sample size was determined using Slovin's formula. Based on the computation, the sample included 61 second-year students, 48 third-year students, and 103 fourth-year students, resulting in a representative sample of the student population.

The formula used is:

$$n = \frac{N}{1 + Ne^2}$$

where:

$$n = \text{sample size}$$

$$N = \text{population size}$$

$$e = \text{margin of error}$$

A total of 10 civil engineering faculty members who were currently teaching or had previously handled the course Statics of Rigid Bodies participated in the study. Participation was voluntary, and respondents were selected according to their willingness and availability during the data collection period. Their responses provided expert evaluation of the developed system in terms of usability, accessibility, and alignment with course content.

Table 3.1: Student Respondents Distribution

Year Level	Population	Sample Size
2nd Year	129	61
3rd Year	102	48
4th Year	218	103
Total	**449**	**212**

3.3 Research Locale

This study was conducted at the College of Engineering of Mindanao State University – General Santos (MSU–Gensan), located in Fatima, General Santos City. The college offers various engineering programs, including Bachelor of Science in Civil Engineering, Electrical Engineering, Electronics and Communication Engineering, Engineering Technology, and Mechanical Engineering.

The research primarily focused on the H-Building, where most classroom lectures, computer laboratory sessions, and faculty offices are located. This setting was considered suitable for the study since it accommodated both the student respondents enrolled in Statics of Rigid Bodies and the instructors teaching the subject.

3.4 Research Procedure

The researchers conceptualized, designed, and developed a web-based learning tool called *StatiCalcs*, an interactive platform with integrated calculators specifically designed for the subject Statics of Rigid Bodies. The system underwent evaluation by academic experts in the field to ensure its accuracy, functionality, and relevance to the course before being introduced to a larger population of users.

A researcher-made questionnaire was developed to measure the level of perception of Civil Engineering students and instructors in terms of usability, accessibility, and satisfaction. The questionnaire was also evaluated by academic experts in the field to ensure its validity, reliability, and appropriateness for the study prior to its administration.

After the evaluation of the questionnaire and the *StatiCalcs* website by academic experts in the field, the researchers proceeded with the implementation of the study. Respondents were first allowed to explore and use the *StatiCalcs* website before the data collection process. They were given approximately fifteen (15) days to use and interact with the platform to ensure adequate exposure and familiarity with its features and tools before answering the survey questionnaire. The researchers then administered the questionnaire to the selected respondents, composed of Bachelor of Science in Civil Engineering students and instructors. After data collection, the gathered responses were encoded, organized, and subjected to statistical analysis.

Participation in the study was voluntary, and all responses were treated with strict confidentiality to ensure the privacy and protection of the respondents.

3.4.1 Website Development Process

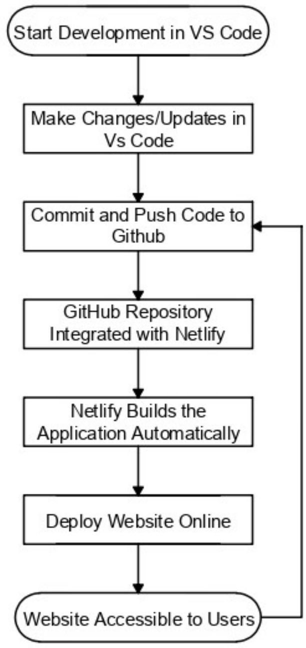

Figure 3.1: System Flow Chart

The website was developed using a modern workflow, beginning with the selection of Java as the backend language due to its reliability, security, and strong support for building large, complex web applications. GitHub was used as the code repository to manage versions, collaborate, and integrate with deployment tools. For the frontend, Tailwind CSS was chosen to create a clean, responsive, and consistent user interface using its utility-first styling approach. ChatGPT assisted throughout the development process by helping with code structuring, debugging, and design decisions, improving efficiency and workflow. Netlify was used as the hosting and deployment platform. By connecting the GitHub repository to Netlify, every update pushed to GitHub was automatically built and deployed online, ensuring continuous delivery and easy accessibility of the website to users.

3.4.2 Website Navigation and Functional Flow

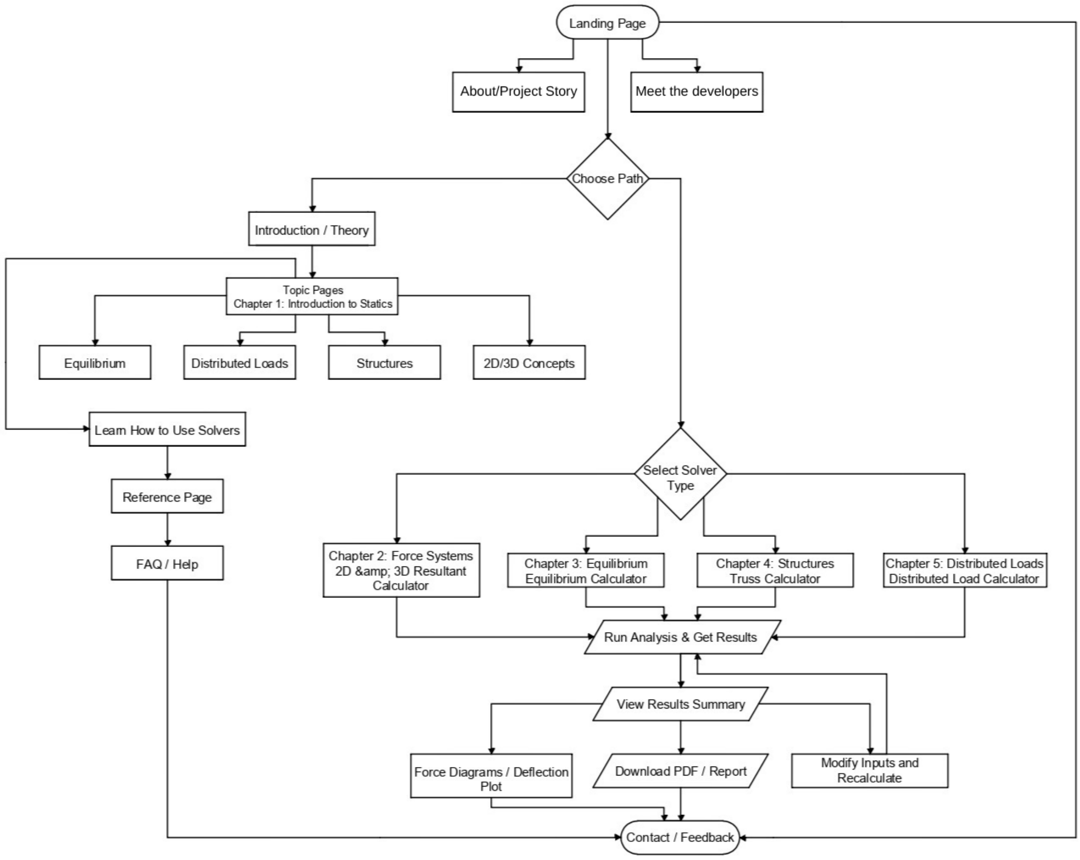

Figure 3.2: User Journey Flow Chart

The website begins at the Landing Page, which serves as the main entry point for all users. From this page, users can navigate to introductory sections such as About / Project Story, Meet the Developers, and Contact / Feedback, which provide background on the project, information about its creators, and a channel for inquiries or suggestions.

From the Landing Page, users are presented with a decision point labeled Choose Path, where they choose between proceeding to the Learning Section or directly accessing the Solver Tools.

If the user selects the Learning Path, they are directed to the Introduction / Theory page, which presents foundational concepts in Statics. From there, users proceed to Topic Pages covering Chapter 1: Modules to Statics, which include lessons on Equilibrium, Distributed Loads, Structures, and 2D/3D Concepts. After engaging with the theoretical content, users are guided to the Learn How to Use Solvers section, which provides instructional guides on properly operating the computational tools. A Reference Page containing formula compilations, example problems, and supplementary materials is also available, alongside an FAQ / Help section for common concerns. Users may also reach the developers directly through the Contact Page.

If the user instead chooses the direct problem-solving pathway, they proceed to a second decision point labeled Select Solver Type. The available solver tools correspond to the chapters covered in the theoretical section:

- Chapter 1: Modules to Statics
- Chapter 2: Force Systems — 2D and 3D Resultant Force Calculator
- Chapter 3: Equilibrium — Concurrent and Non-Concurrent Systems
- Chapter 4: Structures — Truss Analysis Calculator
- Chapter 5: Distributed Loads — Moment of Inertia via Centroidal and Custom Axes

After selecting a solver, the user proceeds to Run Analysis & Get Results. The computed output is then displayed on the View Results Summary page in a clear and structured format. From this page, users have three options:

1. Generate and view Force Diagrams or Deflection Plots
2. Download results as a PDF or formatted report
3. Modify input parameters and Recalculate to explore different solutions or optimize results

This structured flow allows users to either build foundational knowledge before solving problems or directly perform structural analysis depending on their needs, ensuring flexibility, accessibility, and engagement throughout the platform.

3.4.3 Calculator System Flowcharts

The following subsections present the detailed system flowcharts for each calculator integrated into *StatiCalcs*. Each flowchart describes the logical flow of user interactions, input processing, computation steps, and output rendering for its corresponding solver.

2D/3D Resultant Force Calculator

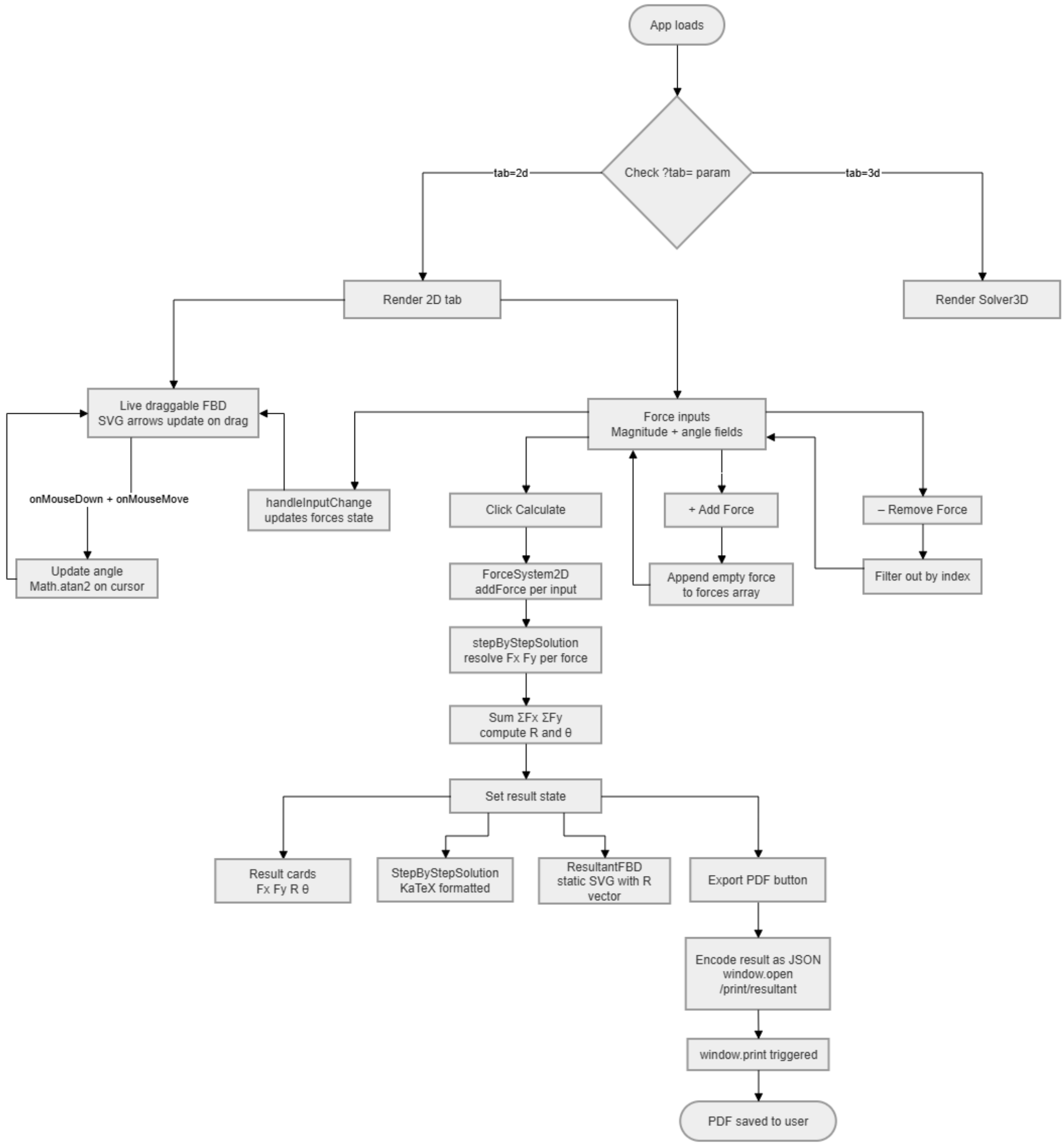

Figure 3.3: 2D/3D Resultant Force Calculator System Flowchart

When the application loads, it checks the URL parameter (`?tab=`) to determine which solver to render. If the parameter is set to `2d`, the 2D tab is rendered; if set to `3d`, the 3D Solver is rendered instead.

In the 2D tab, the user is presented with force input fields for entering the magnitude and angle of each force. Forces can be added by appending an empty entry to the forces array or removed by filtering out by index. As inputs are modified, the `handleInputChange` function updates the forces state in real time. A live draggable Free Body Diagram (FBD) is also displayed, where SVG arrows update dynamically as the user drags them. The angle of each arrow is continuously recalculated using `Math.atan2` based on the cursor position, triggered by `onMouseDown` and `onMouseMove` events.

Once the user clicks Calculate, the system processes each force through the `ForceSystem2D` class using `addForce`.A `stepByStepSolution` is then executed, resolving the x and y components (F_x and F_y) for each force, summing them (ΣF_x and ΣF_y), and computing the resultant force (R) and its direction angle (θ). The computed results are stored in the result state and simultaneously rendered as four outputs: Result Cards displaying F_x, F_y, R, and θ; a Step-by-Step Solution formatted using KaTeX; a static Resultant FBD SVG showing the resultant vector; and an Export PDF Button that encodes the result as JSON, opens the `/print/resultant` route, triggers `window.print`, and saves the output as a PDF for the user.

3D Resultant Force Calculator – Angles Tab

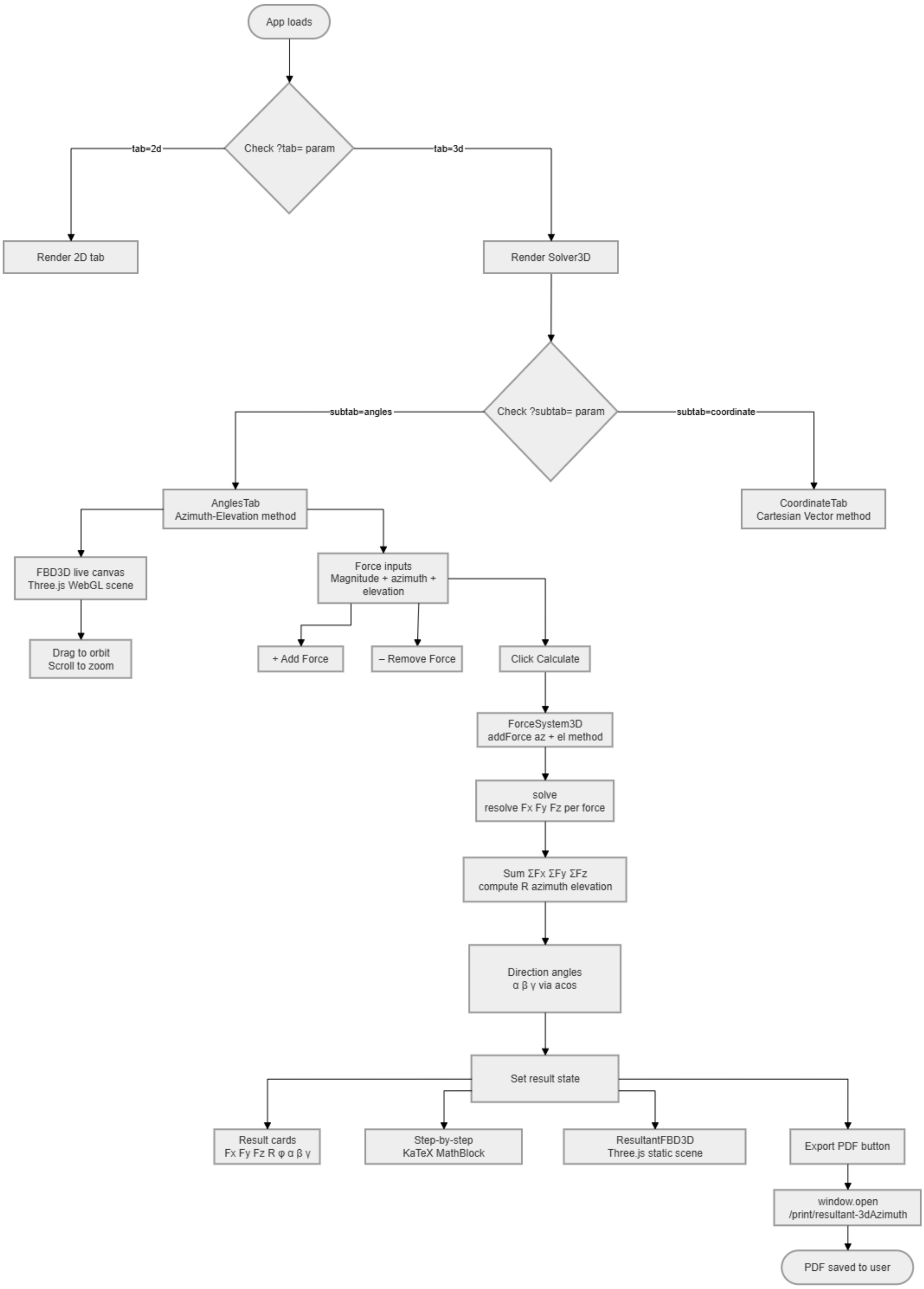

Figure 3.4: 3D Resultant Force Calculator (Angles Tab) System Flowchart

When the application loads and the URL parameter is set to `3d`, the 3D Solver is rendered. A second URL parameter (`?subtab=`) is then checked to determine which method to use. If the subtab is set to `angles`, the AnglesTab is rendered using the Azimuth-Elevation method. If set to `coordinate`, the CoordinateTab is rendered using the Cartesian Vector method instead.

In the Angles Tab, the user is presented with two main components. On the left, a live 3D FBD canvas is rendered using a Three.js WebGL scene, which the user can interact with by dragging to orbit and scrolling to zoom. On the right, the user inputs each force's magnitude, azimuth, and elevation angles. Forces can be added or removed as needed.

Once the user clicks Calculate, each force is processed through the `ForceSystem3D` class using the `addForce` azimuth-elevation method. The system resolves the x, y, and z components (F_x, F_y, F_z) for each force, sums them (ΣF_x, ΣF_y, ΣF_z), and computes the resultant force (R) along with its azimuth and elevation. Direction angles α, β, and γ are then calculated using the `acos` function. The computed results are stored in the result state and simultaneously rendered as four outputs: Result Cards displaying F_x, F_y, F_z, R, φ, α, β, and γ; a Step-by-Step Solution formatted using KaTeX MathBlock; a static Resultant FBD 3D scene rendered using Three.js showing the resultant vector; and an Export PDF Button that opens the `/print/resultant-3dAzimuth` route and saves the output as a PDF for the user.

3D Resultant Force Calculator – Coordinates Tab

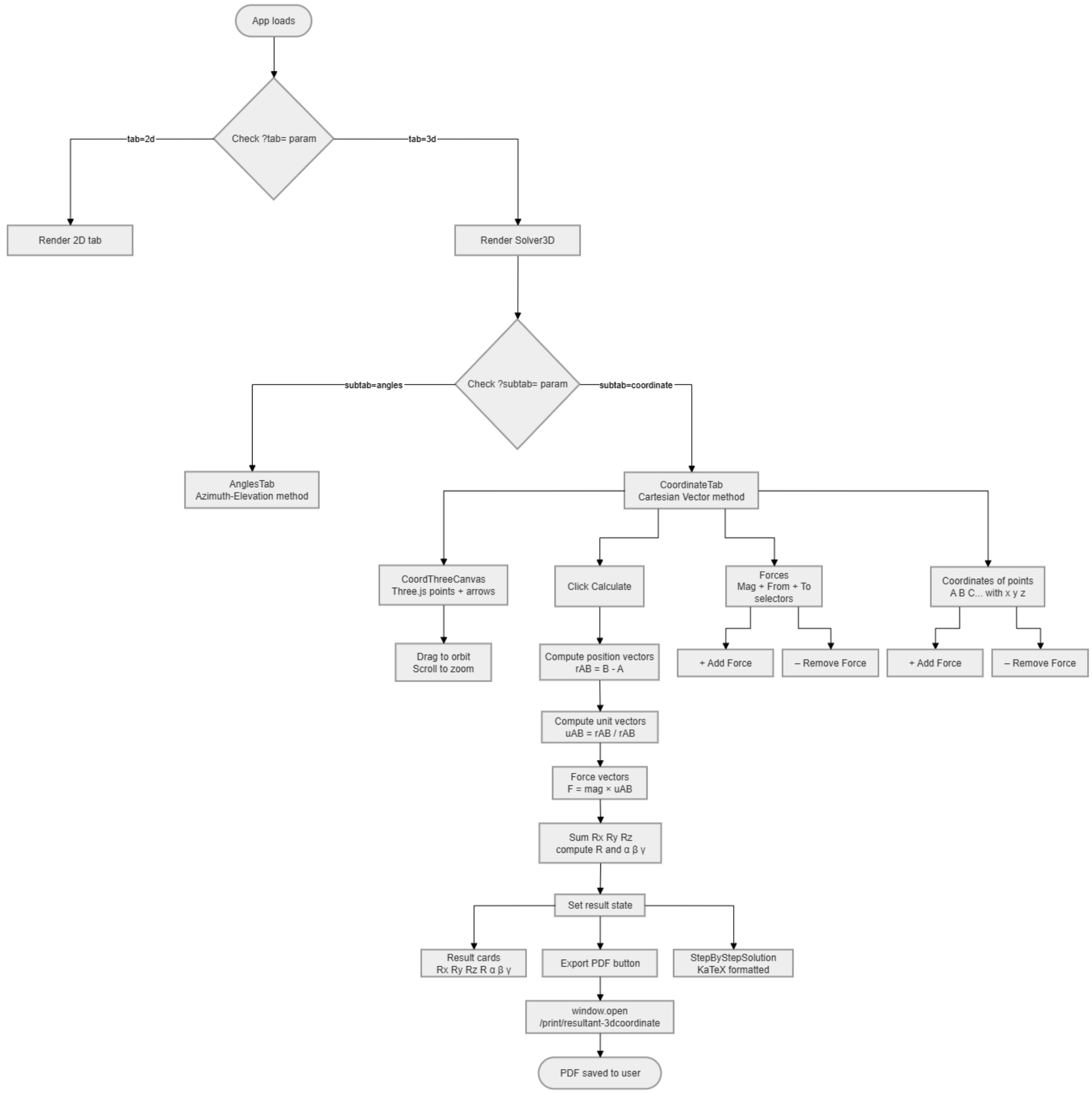

Figure 3.5: 3D Resultant Force Calculator (Coordinates Tab) System Flowchart

If the subtab parameter is set to `coordinate`, the CoordinateTab is rendered using the Cartesian Vector method. The user is presented with three main input areas. On the left, a live 3D canvas is rendered using Three.js, displaying coordinate points and force arrows that the user can interact with by dragging to orbit and scrolling to zoom. In the center, the user manages the force inputs, where each force requires a magnitude and the selection of a *From* and *To* point. On the right, the user defines the coordinates of points (A, B, C, etc.) by entering their x, y, and z values. Forces and coordinate points can be added or removed as needed.

Once the user clicks Calculate, the system processes the inputs through the following steps: position vectors are computed as $\mathbf{r}_{AB} = B - A$; unit vectors are derived as $\mathbf{u}_{AB} = \mathbf{r}_{AB}/|\mathbf{r}_{AB}|$; force vectors are calculated as $\mathbf{F} = \text{magnitude} \times \mathbf{u}_{AB}$; and the components are summed (R_x, R_y, R_z) to compute the resultant force R and the direction angles α, β, and γ. The computed results are stored in the result state and simultaneously rendered as three outputs: Result Cards displaying R_x, R_y, R_z, R, α, β, and γ; a Step-by-Step Solution formatted using KaTeX; and an Export PDF Button that opens the `/print/resultant-3dcoordinate` route and saves the output as a PDF for the user.

Equilibrium of Concurrent Force System

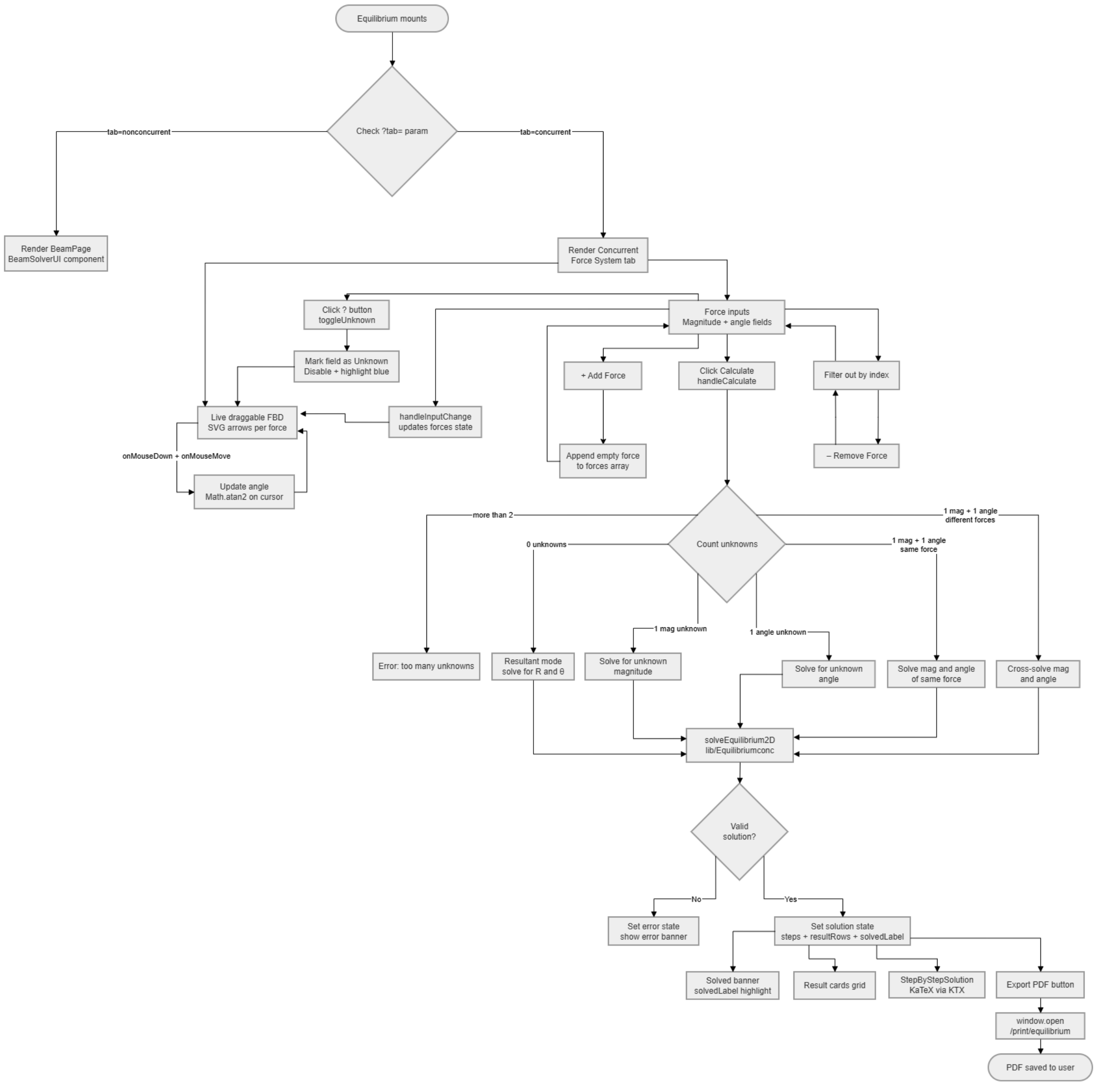

Figure 3.6: Equilibrium of Concurrent Force System Flowchart

When the Equilibrium module loads, it checks the URL parameter (`?tab=`) to determine which solver to render. If set to `concurrent`, the Concurrent Force System tab is rendered. If set to `nonconcurrent`, the BeamPage with the BeamSolverUI component is rendered instead.

In the Concurrent Force System tab, the user is presented with force input fields for magnitude and angle. Forces can be added by appending an empty entry to the forces array or removed by filtering out by index. As inputs are modified, the `handleInputChange` function updates the forces state in real time. The user may also click the `?` button to toggle a field as unknown, which marks and disables that field while highlighting it in blue. A live draggable FBD is displayed alongside the inputs, where SVG arrows update dynamically per force as the user drags, with the angle continuously recalculated using `Math.atan2` based on cursor position via `onMouseDown` and `onMouseMove` events.

Once the user clicks Calculate, the system counts the number of unknowns and branches accordingly:

- More than 2 unknowns → displays an error: too many unknowns
- 0 unknowns → switches to Resultant mode, solving for R and θ
- 1 magnitude unknown → solves for the unknown magnitude
- 1 angle unknown → solves for the unknown angle
- 1 magnitude + 1 angle (same force) → solves for both magnitude and angle of the same force
- 1 magnitude + 1 angle (different forces) → cross-solves for magnitude and angle across different forces

All valid cases are routed through `solveEquilibrium2D` from the `lib/Equilibriumconc` library. The system then checks whether the solution is valid. If invalid, it sets an error state and displays an error banner. If valid, it sets the solution state with steps, result rows, and solved labels, and simultaneously renders four outputs: a Solved Banner with solvedLabel highlight; a Result Cards Grid; a Step-by-Step Solution formatted using KaTeX; and an Export PDF Button that opens the `/print/equilibrium` route and saves the output as a PDF for the user.

Equilibrium of Non-Concurrent Force System – Beam Solver

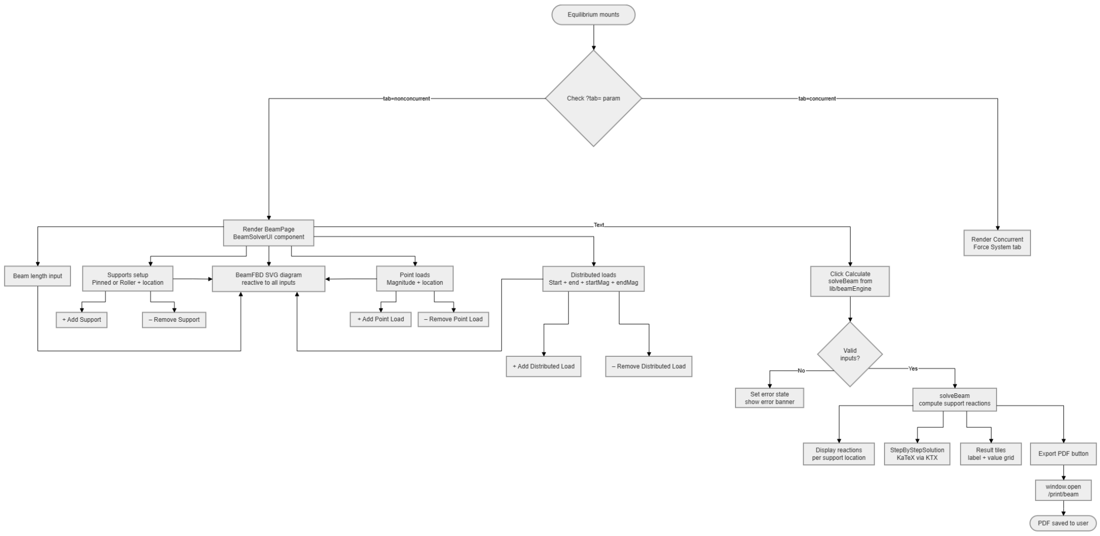

Figure 3.7: Equilibrium of Non-Concurrent Force System (Beam Solver) Flowchart

If the URL parameter is set to `nonconcurrent`, the BeamPage with the BeamSolverUI component is rendered. The user is presented with several input areas that all reactively update a BeamFBD SVG diagram in real time. These include the Beam Length Input, which defines the total length of the beam; the Supports Setup, where the user specifies whether each support is Pinned or Roller along with its location; Point Loads, where the user enters the magnitude and location of each concentrated load; and Distributed Loads, where the user defines the start and end positions along with the start magnitude and end magnitude of each distributed load. All inputs feed into the BeamFBD SVG diagram, which updates reactively to reflect the current beam configuration as the user makes changes.

Once the user clicks Calculate, the system calls `solveBeam` from the `lib/beamEngine` library and checks whether the inputs are valid. If invalid, it sets an error state and displays an error banner. If valid, it executes `solveBeam` to compute the support reactions and simultaneously renders four outputs: Display Reactions per support location; a Step-by-Step Solution formatted using KaTeX; Result Tiles showing a label and value grid; and an Export PDF Button that opens the `/print/beam` route and saves the output as a PDF for the user.

Truss Solver

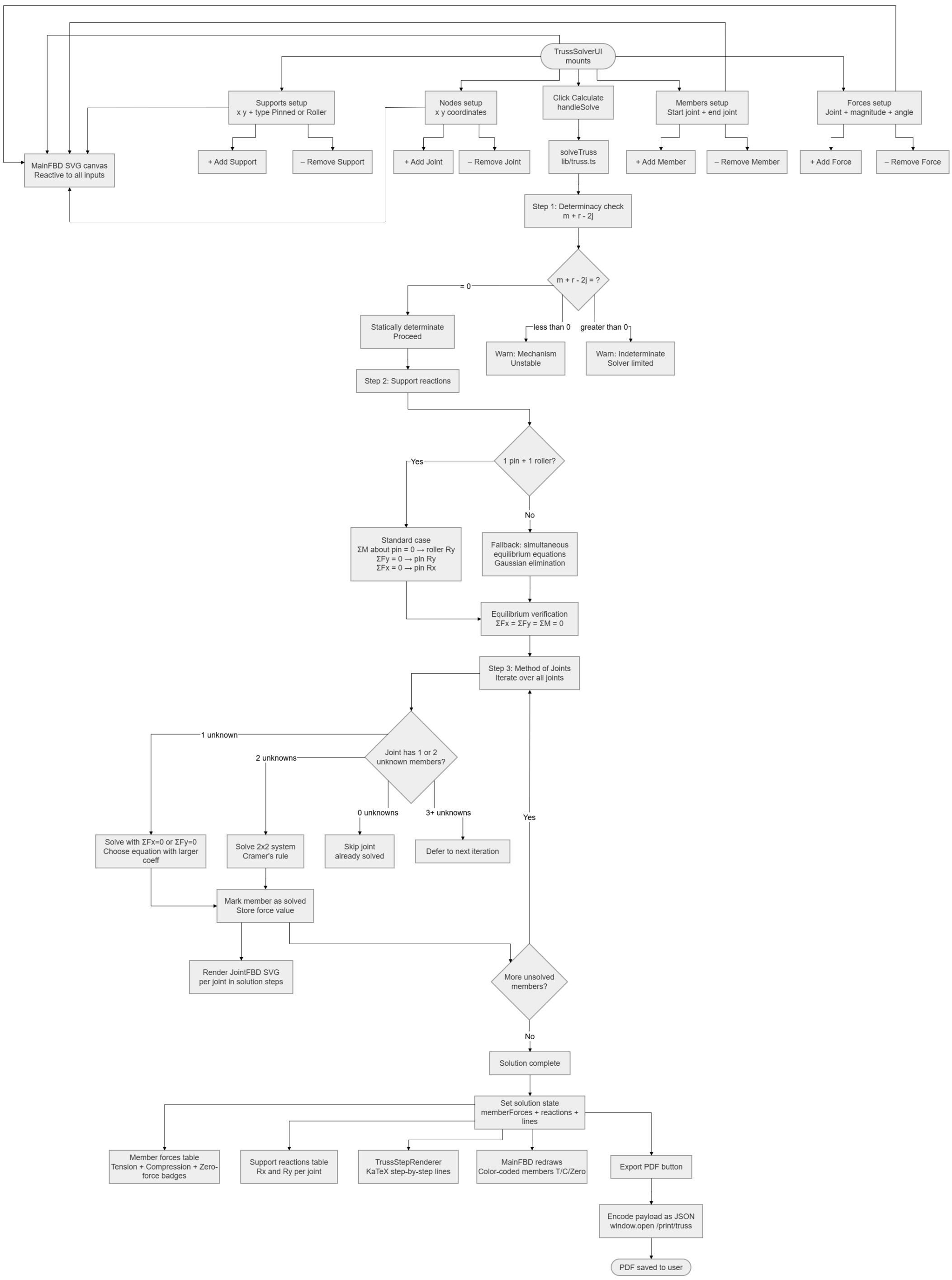

Figure 3.8: Truss Solver System Flowchart

When the TrussSolverUI mounts, the user is presented with four input areas that all reactively update a MainFBD SVG canvas in real time. These include the Supports Setup, where each support is defined by its x, y coordinates and type (Pinned or Roller); Nodes Setup, where joints are added or removed by specifying their x, y coordinates; Members Setup, where each member is defined by selecting a start joint and end joint; and Forces Setup, where external forces are assigned to joints by specifying the joint, magnitude, and angle.

Once the user clicks Calculate, the system calls `solveTruss` from `lib/truss.ts` and proceeds through three major steps.

Step 1: Determinacy Check. The system evaluates the condition $m + r - 2j$. A result of zero indicates a statically determinate truss and the solver proceeds. A result less than zero triggers a warning for mechanism instability, while a result greater than zero triggers a warning that the truss is indeterminate and the solver is limited.

Step 2: Support Reactions. The system checks for the standard configuration of 1 pin and 1 roller. If present, it applies the standard case: ΣM about the pin equals zero for roller R_y, ΣF_y equals zero for pin R_y, and ΣF_x equals zero for pin R_x. Otherwise, it falls back to simultaneous equilibrium equations solved via Gaussian elimination. Equilibrium is then verified by confirming $\Sigma F_x = \Sigma F_y = \Sigma M = 0$.

Step 3: Method of Joints. The system iterates over all joints, checking the number of unknown members at each. Joints with zero unknowns are skipped as already solved. Joints with one unknown are solved using $\Sigma F_x = 0$ or $\Sigma F_y = 0$, selecting the equation with the larger coefficient. Joints with two unknowns are solved using Cramer's rule. Joints with three or more unknowns are deferred to the next iteration. Each solved member is marked and its force value stored, with a JointFBD SVG rendered per joint in the solution steps. The iteration continues until all members are solved and the solution is complete.

The final solution state simultaneously renders five outputs: a Member Forces Table displaying Tension, Compression, and Zero-force badges; a Support Reactions Table showing Rx and Ry per joint; a TrussStepRenderer with KaTeX step-by-step lines; the MainFBD redrawn with color-coded members indicating Tension, Compression, or Zero-force; and an Export PDF Button that encodes the payload as JSON, opens the `/print/truss` route, and saves the output as a PDF for the user.

Moment of Inertia Calculator

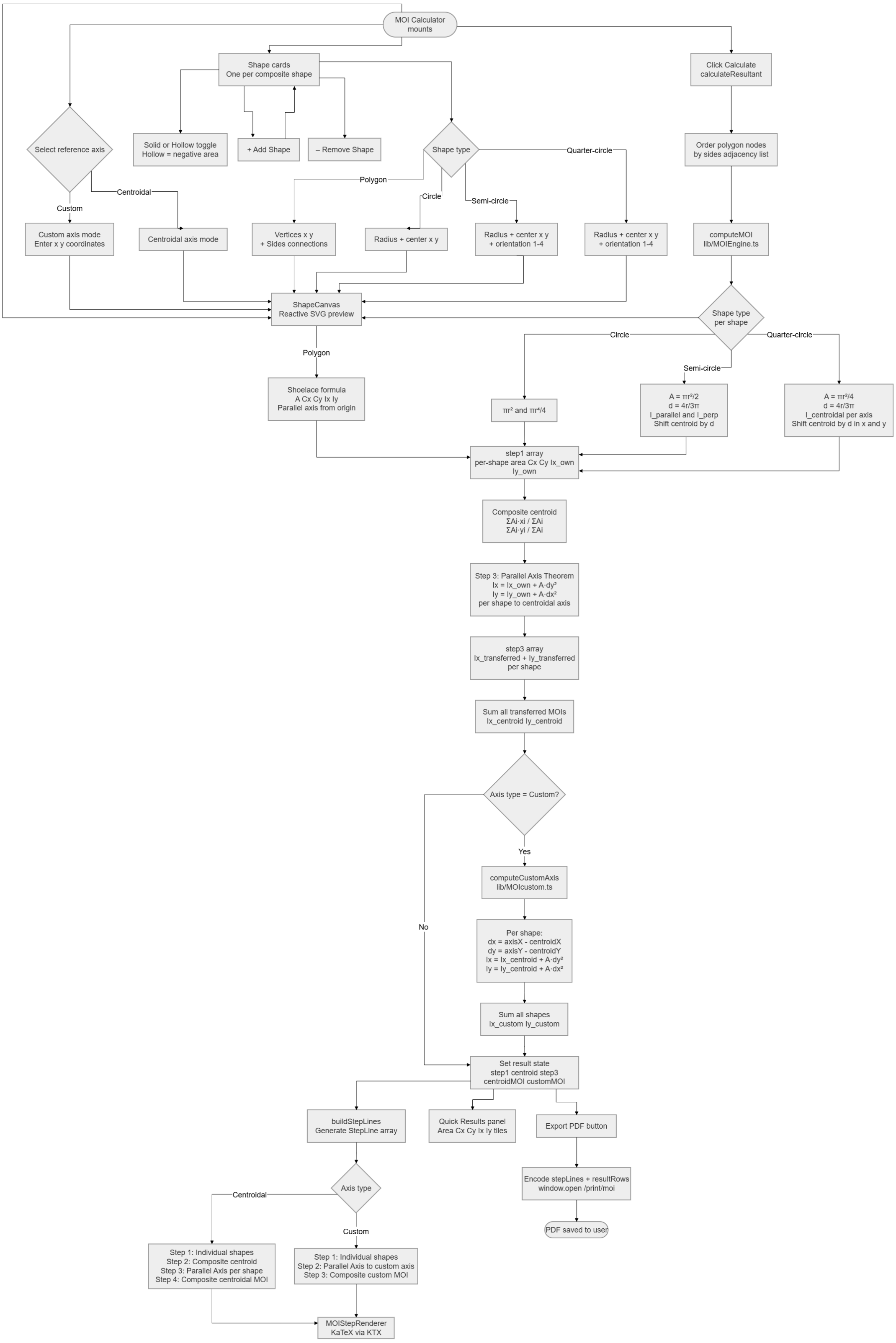

Figure 3.9: Moment of Inertia Calculator System Flowchart

When the MOI Calculator mounts, the user is presented with shape cards, one per composite shape. Shapes can be added or removed, and each shape can be toggled as solid or hollow, where hollow shapes contribute a negative area to the composite section. The user also selects a reference axis, either Centroidal axis mode, which computes the MOI about the composite centroid, or Custom axis mode, which allows the user to enter custom x and y coordinates for the axis. Each shape card requires the selection of a shape type, which determines the required inputs: Polygon requires vertices (x, y) and side connections; Circle requires radius and center (x, y); Semi-circle requires radius, center (x, y), and orientation (1–4); and Quarter-circle requires radius, center (x, y), and orientation (1–4). All inputs feed into a ShapeCanvas, which provides a reactive SVG preview that updates in real time.

Once the user clicks Calculate, the system calls `calculateResultant` from `lib/MOIEngine.ts`. Polygon nodes are ordered by sides adjacency list, and `computeMOI` is called per shape type using the appropriate formulas: Polygon uses the Shoelace formula for area, Cx, Cy, Ix, and Iy with parallel axis from origin; Circle uses πr^2 and $\pi r^4/4$; Semi-circle uses $A = \pi r^2/2$ and $d = 4r/3\pi$ with $I_{parallel}$ and I_{perp} with centroid shifted by d; and Quarter-circle uses $A = \pi r^2/4$ and $d = 4r/3\pi$ with $I_{centroidal}$ per axis and centroid shifted by d in both x and y. The results are assembled into a step1 array containing per-shape area, Cx, Cy, Ix_own, and Iy_own.

The system then proceeds through the following computation steps. The Composite Centroid is computed as $\sum A_i x_i / \sum A_i$ and $\sum A_i y_i / \sum A_i$. The Parallel Axis Theorem is then applied per shape, where $I_x = I_{x,own} + A \cdot dy^2$ and $I_y = I_{y,own} + A \cdot dx^2$, transferring each shape's MOI to the centroidal axis. The transferred values are stored in a step3 array and summed to obtain $I_{x,centroid}$ and $I_{y,centroid}$.

The system then checks whether the axis type is Custom. If yes, it calls `computeCustomAxis` from `lib/MOIcustom.ts`, computing per shape $dx = axisX - centroidX$ and $dy =$

$axisY - centroidY$, then applying the Parallel Axis Theorem again from the centroidal axis to the custom axis, and summing all shapes to obtain $I_{x,custom}$ and $I_{y,custom}$. If no, the system proceeds directly with the centroidal MOI results.

The result state is then set, simultaneously rendering three outputs: a Quick Results Panel showing Area, Cx, Cy, Ix, and Iy tiles; an Export PDF Button that encodes stepLines and resultRows, opens the `/print/moi` route, and saves the output as a PDF for the user; and a MOIStepRenderer formatted using KaTeX. The step-by-step solution is further branched by axis type. For the Centroidal axis, it displays Step 1 covering individual shapes, Step 2 covering the composite centroid, Step 3 covering the parallel axis per shape, and Step 4 covering the composite centroidal MOI. For the Custom axis, it displays Step 1 covering individual shapes, Step 2 covering the parallel axis to the custom axis, and Step 3 covering the composite custom MOI.

3.4.4 Website System Architecture

The *Components* folder contains reusable elements of the website, specifically the Header and Footer, which help maintain consistent navigation and accessibility across all pages. The *Header* file inside this folder includes the main navigation menu with a *Home* button that redirects users to the homepage, a *Calculators* dropdown, a *Modules* dropdown, and a *Take Our Survey* button that directs users to the external survey form.

The *Calculators* dropdown provides quick access to all integrated solver tools, including the 2D Resultant Force Calculator, the 3D Resultant Force Calculator via Cartesian Vector method, the 3D Resultant Force Calculator via Azimuth-Elevation method, the Concurrent Unknown Forces and Angles Calculator, the Non-Concurrent Beam Analysis solver, the Truss Analysis Calculator, and the Moment of Inertia Calculator. The *Modules* dropdown provides access to eight learning modules, namely Module 1: Free Body Diagram, Module 2: 2D Resultant, Module 3: 3D Resultant, Module 4: 2D Equilibrium, Module 5: 3D Equilibrium, Module 6: Truss Joint Method, Module 7: MOI Centroidal Axis, and Module 8: MOI Custom Axis. These dropdowns are designed to make browsing easier and more efficient for users.

Meanwhile, the Footer file contains additional navigation links including *About*, *References*, and *Developers*. The About button directs users to the website's About page; the References page provides the sources used in developing the calculators and learning materials; and the Developers page displays information about the creators of the website. These components together ensure that users can easily access important sections and navigate the site smoothly.

The *Calcs* folder contains the core Java code files that power the calculators integrated into the website. These files are responsible for performing the computational logic, processing user inputs, and generating the corresponding solutions based on engineering principles and formulas. Each calculator, such as those for force systems, equilibrium, truss analysis, and distributed loads, relies on the scripts stored in this folder to accurately compute results. This makes the Calcs folder an essential component of the website's functionality, as it handles the main calculation processes that support the learning and problem-solving features of the platform.

The *App* folder contains all the front-end code for all website pages, styled using Tailwind CSS, which controls their visual layout, responsiveness, and overall user interface. Each page inside the App folder is responsible for either displaying calculators, such as for 2D Resultant, 3D Solver, Equilibrium, Distributed Loads, and Structural Analysis, or for providing supporting information, such as Introduction, References, About, Contact, and Developer details. The Tailwind CSS classes ensure that all these pages are consistently designed, user-friendly, and visually appealing, making navigation smooth and calculator usage clear and efficient. In summary, the App folder manages the appearance, layout, and behavior of both the calculator and informational pages from the user's perspective.

3.5 Research Instrument

This study utilized a researcher-made questionnaire to evaluate the respondents' perceptions of the developed web-based learning tool, *StatiCalcs*. The questionnaire focused on three areas: usability, accessibility, and satisfaction. Each area contained five (5) questions, resulting in a total of fifteen (15) statements. Responses were measured using a five-point Likert scale ranging from 1 (Strongly Disagree) to 5 (Strongly Agree). The responses gathered provided quantitative data regarding the respondents' perceptions of the platform.

In addition, the questionnaire included a section that determined the respondents' level of preference for different learning resources when studying Statics. This section contained six (6) statements measured using a five-point preference scale ranging from 1 (Least Preferred) to 5 (Most Preferred). The learning resources included textbooks, modules, YouTube videos, lecture notes, online engineering tools, and *StatiCalcs*.

The questionnaire also included one (1) open-ended question that allowed respondents to provide comments, suggestions, or feedback that may help improve the *StatiCalcs* platform.

To ensure the validity and reliability of the instrument, the questionnaire was evaluated by academic experts in the field prior to its administration.

Table 3.2: Likert Scale for Respondent Perception

Scale	**Range**	**Description**	**Interpretation**
5	4.21–5.00	Strongly Agree	The respondent fully agrees with the statement.
4	3.41–4.20	Agree	The respondent generally agrees with the statement.
3	2.61–3.40	Neutral	The respondent neither agrees nor disagrees with the statement.
2	1.81–2.60	Disagree	The respondent generally disagrees with the statement.
1	1.00–1.80	Strongly Disagree	The respondent fully disagrees with the statement.

3.6 Statistical Tools

Weighted Mean

The weighted mean was used to determine the respondents' level of perception of the *StatiCalcs* platform in terms of usability, accessibility, and satisfaction. It was also used to determine the respondents' level of preference for different learning resources in studying Statics.

The following formula was used to compute the weighted mean:

$$\bar{X} = \frac{\sum fx}{N}$$

where:

- $\bar{X}$ = weighted mean

- f = frequency of responses
- x = weight assigned to each response
- N = total number of responses

Cronbach's Alpha

Cronbach's alpha was used to determine the internal consistency and reliability of the questionnaire. Separate reliability tests were conducted for usability, accessibility, satisfaction, and preference for learning resources.

The following formula was used to compute Cronbach's alpha:

$$\alpha = \frac{k}{k-1}\left(1 - \frac{\sum \sigma_i^2}{\sigma_t^2}\right)$$

where:

- k = number of items
- σ_i^2 = variance of each item
- σ_t^2 = total variance

The interpretation of Cronbach's alpha is presented in Table X.

Table 3.3: Interpretation of Cronbach's Alpha

Cronbach's Alpha	**Interpretation**
0.90 – 1.00	Excellent
0.80 – 0.89	Good
0.70 – 0.79	Acceptable
0.60 – 0.69	Questionable
Below 0.60	Poor

The computed Cronbach's alpha values are presented in Table X.

Table 3.4: Computed Cronbach's Alpha Values

Variable	**No. of Items**	**Cronbach's Alpha**	**Interpretation**
Usability	5	0.88	Good
Accessibility	5	0.94	Excellent
Satisfaction	5	0.94	Excellent
Preference for Learning Resources	6	0.86	Good

The computed Cronbach's alpha values indicated that the questionnaire demonstrated acceptable internal consistency and reliability for use in the study.

Independent Samples t-test

An independent samples t-test was used to determine whether there was a significant difference between the perceptions of Civil Engineering students and instructors regarding the usability, accessibility, and satisfaction of the *StatiCalcs* platform.

The following formula was used to compute the independent samples t-test:

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

where:

- t = computed t-value
- $\bar{X}_1$ = mean of the first group
- $\bar{X}_2$ = mean of the second group
- s_1^2 = variance of the first group
- s_2^2 = variance of the second group

- n_1 = number of respondents in the first group
- n_2 = number of respondents in the second group

The independent samples t-test was conducted at a 0.05 level of significance.

3.7 Statistical Treatment

This study utilized weighted mean to analyze the respondents' perceptions of the developed web-based learning tool, *StatiCalcs*, in terms of usability, accessibility, and satisfaction, as well as their level of preference for different learning resources in studying Statics. Each item in the questionnaire was measured using a five-point scale. The weighted mean was used to determine the overall responses of the respondents for each evaluated area.

In addition, an independent samples t-test was conducted to determine whether there was a significant difference between the perceptions of Civil Engineering students and instructors regarding the usability, accessibility, and satisfaction of the *StatiCalcs* platform. The test was conducted using a 0.05 level of significance.

Chapter 4
Results and Discussions

This chapter presents the analysis and interpretation of the data gathered from the respondents regarding the developed web-based learning tool, *StatiCalcs*. The findings focus on the respondents' perceptions in terms of usability, accessibility, and satisfaction, as well as their preference for different learning resources in studying Statics of Rigid Bodies. Frequency distribution, weighted mean, and independent samples t-test were utilized to analyze the gathered data.

4.1 Level of Perception of Students Regarding StatiCalcs in Terms of Usability

Table 4.1: Frequency Distribution of Students' Responses on Usability

Statement	**SA**	**A**	**N**	**D**	**SD**
The StatiCalcs website is easy to navigate.	155	48	0	0	0
I can complete calculations quickly using the website.	140	54	0	2	0
The website provides clear feedback after performing calculations.	150	52	0	1	0
I can easily correct mistakes when entering inputs in the calculators.	159	42	0	0	0
Overall, StatiCalcs is user-friendly.	160	43	0	0	0

Table 4.2: Weighted Mean of Students' Perception on Usability

Statement	**Weighted Mean**	**Interpretation**
The StatiCalcs website is easy to navigate.	4.69	Strongly Agree
I can complete calculations quickly using the website.	4.57	Strongly Agree
The website provides clear feedback after performing calculations.	4.66	Strongly Agree
I can easily correct mistakes when entering inputs in the calculators.	4.70	Strongly Agree
Overall, StatiCalcs is user-friendly.	4.71	Strongly Agree
Overall Weighted Mean	**4.66**	**Strongly Agree**

The findings reveal that the student respondents strongly agreed that *StatiCalcs* is usable and user-friendly. The statement, "Overall, StatiCalcs is user-friendly," obtained the highest weighted mean of 4.71, indicating a highly positive evaluation of the platform's overall usability. Meanwhile, the statement, "I can complete calculations quickly using the website," obtained the lowest weighted mean of 4.57. Despite this, all indicators remained within the "Strongly Agree" interpretation range. The results suggest that the respondents perceived the developed web-based learning tool as easy to navigate, efficient, and beneficial for solving Statics problems.

4.2 Level of Perception of Students Regarding StatiCalcs in Terms of Accessibility

Table 4.3: Frequency Distribution of Students' Responses on Accessibility

Statement	SA	A	N	D	SD
The text and visual elements of StatiCalcs are easy to read.	146	59	0	0	0
The website works properly on different devices such as laptops and mobile phones.	150	52	0	1	0
The layout makes important information easy to locate.	154	48	0	0	0
The explanations and results provided by the website are easy to understand.	144	51	0	0	0
Overall, StatiCalcs is accessible under different conditions.	142	60	0	0	0

Table 4.4: Weighted Mean of Students' Perception on Accessibility

Statement	Weighted Mean	Interpretation
The text and visual elements of StatiCalcs are easy to read.	4.66	Strongly Agree
The website works properly on different devices such as laptops and mobile phones.	4.66	Strongly Agree
The layout makes important information easy to locate.	4.69	Strongly Agree
The explanations and results provided by the website are easy to understand.	4.61	Strongly Agree
Overall, StatiCalcs is accessible under different conditions.	4.63	Strongly Agree
Overall Weighted Mean	**4.65**	**Strongly Agree**

The results indicate that the student respondents strongly agreed that *StatiCalcs* is accessible and functional across different devices and learning conditions. The statement, "The layout makes important information easy to locate," obtained the highest weighted mean of 4.69, suggesting that the organization and presentation of information in the platform supported ease of use and navigation. On the other hand, the statement, "The explanations and results provided by the website are easy to understand," obtained the lowest weighted mean of 4.61. Nevertheless, all indicators were interpreted as "Strongly Agree," indicating that the respondents positively evaluated the accessibility features of the developed platform.

4.3 Level of Perception of Students Regarding StatiCalcs in Terms of Satisfaction

Table 4.5: Frequency Distribution of Students' Responses on Satisfaction

Statement	**SA**	**A**	**N**	**D**	**SD**
I am satisfied with my overall experience using StatiCalcs.	157	44	0	0	0
StatiCalcs meets my expectations as a supplementary learning tool.	155	44	0	0	0
The features of StatiCalcs are beneficial for practice and problem solving.	170	33	0	0	0
I trust the accuracy of the calculations generated by StatiCalcs.	140	50	0	0	0
Compared to textbooks, modules, and online videos, StatiCalcs provides a more convenient learning experience.	133	46	0	2	0

Table 4.6: Weighted Mean of Students' Perception on Satisfaction

Statement	**Weighted Mean**	**Interpretation**
I am satisfied with my overall experience using StatiCalcs.	4.69	Strongly Agree
StatiCalcs meets my expectations as a supplementary learning tool.	4.67	Strongly Agree
The features of StatiCalcs are beneficial for practice and problem solving.	4.76	Strongly Agree
I trust the accuracy of the calculations generated by StatiCalcs.	4.56	Strongly Agree
Compared to textbooks, modules, and online videos, StatiCalcs provides a more convenient learning experience.	4.46	Strongly Agree
Overall Weighted Mean	**4.63**	**Strongly Agree**

The findings reveal that the student respondents were highly satisfied with the developed *StatiCalcs* platform. The statement, "The features of StatiCalcs are beneficial for practice and problem solving," obtained the highest weighted mean of 4.76, indicating that the respondents recognized the usefulness of the platform in solving engineering problems and reinforcing learning. Meanwhile, the statement comparing *StatiCalcs* with traditional learning resources obtained the lowest weighted mean of 4.46. However, all indicators remained within the "Strongly Agree" interpretation range, suggesting that the respondents perceived *StatiCalcs* as an effective and satisfactory supplementary learning tool for studying Statics of Rigid Bodies.

4.4 Level of Preference of Students for Different Learning Resources in Studying Statics of Rigid Bodies

Table 4.7: Frequency Distribution of Students' Preference for Learning Resources

Learning Resource	**MP**	**P**	**N**	**LP**	**LEP**
Textbooks	128	68	14	2	0
Modules	99	83	28	2	0
YouTube Videos	107	73	28	3	1
Lecture Notes	58	95	55	4	0
Online Engineering Tools	88	50	62	10	2
StatiCalcs	129	73	8	2	0

Table 4.8: Weighted Mean of Students' Preference for Learning Resources

Learning Resource	**Weighted Mean**	**Interpretation**
Textbooks	4.52	Most Preferred
Modules	4.32	Most Preferred
YouTube Videos	4.33	Most Preferred
Lecture Notes	3.98	Preferred
Online Engineering Tools	4.00	Preferred
StatiCalcs	4.55	Most Preferred

The findings reveal that *StatiCalcs* was the most preferred learning resource among the student respondents, obtaining the highest weighted mean of 4.55, interpreted as "Most Preferred." This was closely followed by textbooks, which obtained a weighted mean of 4.52. The results suggest that while students still value traditional learning resources, they also demonstrate strong acceptance of technology-supported learning tools for studying Statics of Rigid Bodies. Meanwhile, lecture notes obtained the lowest weighted mean of 3.98, interpreted as "Preferred," indicating that students tend to favor more interactive and comprehensive learning resources.

4.5 Level of Perception of Instructors Regarding StatiCalcs in Terms of Usability

Table 4.9: Frequency Distribution of Instructors' Responses on Usability

Statement	**SA**	**A**	**N**	**D**	**SD**
The StatiCalcs website is easy to navigate.	6	2	0	0	0
I can complete calculations quickly using the website.	5	3	0	0	0
The website provides clear feedback after performing calculations.	3	4	0	0	0
I can easily correct mistakes when entering inputs in the calculators.	5	2	0	0	0
Overall, StatiCalcs is user-friendly.	5	3	0	0	0

Table 4.10: Weighted Mean of Instructors' Perception on Usability

Statement	**Weighted Mean**	**Interpretation**
The StatiCalcs website is easy to navigate.	4.75	Strongly Agree
I can complete calculations quickly using the website.	4.63	Strongly Agree
The website provides clear feedback after performing calculations.	4.25	Strongly Agree
I can easily correct mistakes when entering inputs in the calculators.	4.50	Strongly Agree
Overall, StatiCalcs is user-friendly.	4.63	Strongly Agree
Overall Weighted Mean	**4.55**	**Strongly Agree**

The results indicate that the instructor respondents strongly agreed that *StatiCalcs* is usable and easy to navigate. The statement, "The StatiCalcs website is easy to navigate," obtained the highest weighted mean of 4.75, indicating a highly favorable evaluation of the platform's interface and usability. Meanwhile, the statement, "The website provides clear feedback after performing calculations," obtained the lowest weighted mean of 4.25. Nevertheless, all indicators were interpreted as "Strongly Agree," demonstrating that the instructors positively perceived the usability of the developed learning platform.

4.6 Level of Perception of Instructors Regarding StatiCalcs in Terms of Accessibility

Table 4.11: Frequency Distribution of Instructors' Responses on Accessibility

Statement	SA	A	N	D	SD
The text and visual elements of StatiCalcs are easy to read.	5	2	1	0	0
The website works properly on different devices such as laptops and mobile phones.	5	3	0	0	0
The layout makes important information easy to locate.	4	4	0	0	0
The explanations and results provided by the website are easy to understand.	5	3	0	0	0
Overall, StatiCalcs is accessible under different conditions.	2	5	1	0	0

Table 4.12: Weighted Mean of Instructors' Perception on Accessibility

Statement	**Weighted Mean**	**Interpretation**
The text and visual elements of StatiCalcs are easy to read.	4.50	Strongly Agree
The website works properly on different devices such as laptops and mobile phones.	4.63	Strongly Agree
The layout makes important information easy to locate.	4.50	Strongly Agree
The explanations and results provided by the website are easy to understand.	4.63	Strongly Agree
Overall, StatiCalcs is accessible under different conditions.	4.13	Agree
Overall Weighted Mean	**4.48**	**Strongly Agree**

The findings indicate that the instructor respondents generally perceived *StatiCalcs* as accessible and functional across different conditions. The statements, "The website works properly on different devices such as laptops and mobile phones" and "The explanations and results provided by the website are easy to understand," obtained the highest weighted means of 4.63. Meanwhile, the statement, "Overall, StatiCalcs is accessible under different conditions," obtained the lowest weighted mean of 4.13, interpreted as "Agree." Despite this, the overall weighted mean of 4.48 indicates that the instructors positively evaluated the accessibility of the developed web-based learning tool.

4.7 Level of Perception of Instructors Regarding StatiCalcs in Terms of Satisfaction

Table 4.13: Frequency Distribution of Instructors' Responses on Satisfaction

Statement	SA	A	N	D	SD
I am satisfied with my overall experience using StatiCalcs.	4	4	0	0	0
StatiCalcs meets my expectations as a supplementary learning tool.	4	4	0	0	0
The features of StatiCalcs are beneficial for practice and problem solving.	5	2	0	0	0
I trust the accuracy of the calculations generated by StatiCalcs.	5	3	0	0	0
Compared to textbooks, modules, and online videos, StatiCalcs provides a more convenient learning experience.	2	3	3	0	0

Table 4.14: Weighted Mean of Instructors' Perception on Satisfaction

Statement	**Weighted Mean**	**Interpretation**
I am satisfied with my overall experience using StatiCalcs.	4.50	Strongly Agree
StatiCalcs meets my expectations as a supplementary learning tool.	4.50	Strongly Agree
The features of StatiCalcs are beneficial for practice and problem solving.	4.50	Strongly Agree
I trust the accuracy of the calculations generated by StatiCalcs.	4.63	Strongly Agree
Compared to textbooks, modules, and online videos, StatiCalcs provides a more convenient learning experience.	3.88	Agree
Overall Weighted Mean	**4.40**	**Strongly Agree**

The findings reveal that the instructor respondents were generally satisfied with *StatiCalcs* as a supplementary learning tool. The statement, "I trust the accuracy of the calculations generated by StatiCalcs," obtained the highest weighted mean of 4.63, indicating confidence in the correctness and reliability of the platform's outputs. Meanwhile, the statement comparing *StatiCalcs* with textbooks, modules, and online videos obtained the lowest weighted mean of 3.88, interpreted as "Agree." Nevertheless, the overall results indicate that instructors positively evaluated the developed platform in terms of satisfaction.

4.8 Level of Preference of Instructors for Different Learning Resources in Studying Statics of Rigid Bodies

Table 4.15: Frequency Distribution of Instructors' Preference for Learning Resources

Learning Resource	**MP**	**P**	**N**	**LP**	**LEP**
Textbooks	6	1	0	1	0
Modules	0	6	2	0	0
YouTube Videos	0	3	3	1	1
Lecture Notes	0	7	1	0	0
Online Engineering Tools	1	4	3	0	0
StatiCalcs	0	8	0	0	0

Table 4.16: Weighted Mean of Instructors' Preference for Learning Resources

Learning Resource	**Weighted Mean**	**Interpretation**
Textbooks	4.50	Most Preferred
Modules	3.75	Preferred
YouTube Videos	3.00	Neutral
Lecture Notes	3.88	Preferred
Online Engineering Tools	3.75	Preferred
StatiCalcs	4.00	Preferred

The results indicate that the instructor respondents still preferred traditional learning resources, particularly textbooks, which obtained the highest weighted mean of 4.50, interpreted as "Most Preferred." *StatiCalcs* obtained a weighted mean of 4.00, interpreted as "Preferred," indicating positive acceptance of the developed web-based learning tool among instructors. Meanwhile, YouTube videos obtained the lowest weighted mean of 3.00, interpreted as "Neutral." The findings suggest that instructors continue to value conventional academic resources while recognizing the usefulness of technology-supported learning tools.

4.9 Significant Difference Between the Perceptions of Students and Instructors Regarding StatiCalcs

Table 4.17: Test of Significant Difference Between the Perceptions of Students and Instructors

Variable	**Student Mean**	**Instructor Mean**	**P-value**	**Decision**	**Interpretation**
Usability	4.66	4.55	0.55	Accept Ho	No Significant Difference
Accessibility	4.65	4.48	0.62	Accept Ho	No Significant Difference
Satisfaction	4.63	4.40	0.27	Accept Ho	No Significant Difference

The student respondents obtained weighted mean values of 4.66 for usability, 4.65 for accessibility, and 4.63 for satisfaction. These results indicate that the students positively perceived *StatiCalcs* as a usable, accessible, and satisfactory web-based learning tool for studying Statics of Rigid Bodies. Among the evaluated variables, usability obtained the highest weighted mean, suggesting that the respondents found the platform easy to navigate and user-friendly.

On the other hand, the instructor respondents obtained weighted mean values of 4.55

for usability, 4.48 for accessibility, and 4.40 for satisfaction. The results likewise reflect positive perceptions toward the developed learning tool. Furthermore, all computed p-values were greater than the 0.05 level of significance; therefore, the null hypothesis was accepted. This indicates that there was no significant difference between the perceptions of students and instructors regarding the usability, accessibility, and satisfaction of *StatiCalcs*.

4.10 Respondents' Comments and Suggestions Regarding StatiCalcs

The respondents provided several comments, suggestions, and observations regarding the developed *StatiCalcs* platform. Most responses expressed positive perceptions toward the usability, accessibility, and usefulness of the website as a supplementary learning tool for studying Statics of Rigid Bodies. Several respondents described the platform as user-friendly, easy to navigate, helpful, convenient, and effective for checking answers, reviewing lessons, and understanding engineering concepts. Many respondents also appreciated the integration of calculators, modules, formulas, and learning materials within a single platform. Some respondents expressed that the website would be highly beneficial for engineering students who struggle with Statics and related subjects.

Despite the positive feedback, respondents also provided several recommendations for future improvements of the platform. Common suggestions included the addition of step-by-step solutions, more examples and practice problems, formula displays, diagrams, and video tutorials to improve learning support and usability. Other respondents recommended improving navigation features by adding back and next buttons, enhancing formula visibility, implementing input validation, and strengthening mobile accessibility. Some respondents also suggested expanding the scope of the platform by incorporating additional engineering topics such as Dynamics of Rigid Bodies and Mechanics of Deformable Bodies, supporting multiple units of measurement, improving calculator functionality, and adding visualization features for better conceptual understanding. Overall, the comments and suggestions indicate that the respondents positively received the platform and recognized its strong potential for further enhancement and broader educational application.

Chapter 5
Summary, Conclusion, and Recommendations

5.1 Summary of Findings

5.2 Conclusions

5.3 Recommendations

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Appendix A

Research Questionnaire

Instructions

Please read each statement carefully. Put a check mark on the option that best describes your perception of the developed web-based learning tool, **StatiCalcs**. There are no right or wrong answers. All responses will be treated with strict confidentiality.

Part I. Respondent Information

Name (Optional): ____

Status (Please check one): ☐ Student ☐ Instructor

Year Level (For students only): ____

Have you used the StatiCalcs website? ☐ Yes ☐ No

Part II. Usability

This section evaluates how easy and efficient the StatiCalcs platform is to use.

Range	**Description**	**Interpretation**
4.21 – 5.00	Strongly Agree	I fully agree with the statement.
3.41 – 4.20	Agree	I generally agree with the statement.
2.61 – 3.40	Neutral	I neither agree nor disagree with the statement.
1.81 – 2.60	Disagree	I generally disagree with the statement.
1.00 – 1.80	Strongly Disagree	I fully disagree with the statement.

No.	Statement	5	4	3	2	1
1	The StatiCalcs website is easy to navigate.	☐	☐	☐	☐	☐
2	I can complete calculations quickly using the website.	☐	☐	☐	☐	☐
3	The website provides clear feedback after performing calculations.	☐	☐	☐	☐	☐
4	I can easily correct mistakes when entering inputs in the calculators.	☐	☐	☐	☐	☐
5	Overall, StatiCalcs is user-friendly.	☐	☐	☐	☐	☐

Part III. Accessibility

This section evaluates how easily the StatiCalcs platform can be accessed.

No.	Statement	5	4	3	2	1
1	The text and visual elements of StatiCalcs are easy to read.	☐	☐	☐	☐	☐
2	The website works properly on different devices such as laptops and mobile phones.	☐	☐	☐	☐	☐
3	The layout makes important information easy to locate.	☐	☐	☐	☐	☐
4	The explanations and results provided by the website are easy to understand.	☐	☐	☐	☐	☐
5	Overall, StatiCalcs is accessible under different conditions.	☐	☐	☐	☐	☐

Part IV. User Satisfaction

This section evaluates overall satisfaction with the StatiCalcs platform.

No.	**Statement**	**5**	**4**	**3**	**2**	**1**
1	I am satisfied with my overall experience using StatiCalcs.	☐	☐	☐	☐	☐
2	StatiCalcs meets my expectations as a supplementary learning tool.	☐	☐	☐	☐	☐
3	The features of StatiCalcs are beneficial for practice and problem solving.	☐	☐	☐	☐	☐
4	I trust the accuracy of the calculations generated by StatiCalcs.	☐	☐	☐	☐	☐
5	Compared to textbooks, modules, and online videos, StatiCalcs provides a more convenient learning experience.	☐	☐	☐	☐	☐

Part V. Preference of Learning Resources

This section determines the respondents' level of preference for different learning resources when studying Statics.

Scale	Description	Interpretation
5	Most Preferred	I strongly prefer using the learning resource.
4	Preferred	I generally prefer using the learning resource.
3	Neutral	I have no strong preference for the learning resource.
2	Less Preferred	I generally prefer other learning resources over this.
1	Least Preferred	I strongly prefer other learning resources over this.

No.	Learning Resource	5	4	3	2	1
1	Textbooks (e.g., Hibbeler, Beer and Johnston)	☐	☐	☐	☐	☐
2	Modules (e.g., instructor-provided materials)	☐	☐	☐	☐	☐
3	YouTube Videos (e.g., educational tutorials)	☐	☐	☐	☐	☐
4	Lecture Notes (e.g., class notes, handouts)	☐	☐	☐	☐	☐
5	Online Engineering Tools (e.g., SkyCiv, Mathalino)	☐	☐	☐	☐	☐
6	StatiCalcs (web-based learning tool)	☐	☐	☐	☐	☐

Part VI. Open-Ended Question

Please share any comments, suggestions, or feedback that may help improve the StatiCalcs platform.

Appendix B

Homepage

```
import Link from "next/link";
import Header from "../components/Header";
import Footer from "../components/Footer";

export default function HomePage(){
  return(
    <main className="bg-gray-50 min-h-screen flex flex-col relative">
      <div/>
      <Header/>
      <section className="flex flex-col items-center text-center py-12
      bg-gray-50 flex-grow">
        <h1 className="text-[20px] md:text-[80px] font-bold mb-2
        text-black">
          Stati<span className="text-[#1848a0]">Calcs</span>
        </h1>
        <p className="text-gray-600 mb-12 text-[18px]">
          Interactive calculators for learning and solving Statics of
          Rigid Bodies.
        </p>
        <div className="w-full max-w-2xl">
          <div className="grid grid-cols-2 gap-4 items-center">
            <p className="text-left text-black text-[18px]
            font-bold">Chapter 1: Modules to Statics</p>
            <Link href="/Introduction" className="bg-[#1848a0]
            text-white px-6 py-3 rounded-md shadow hover:bg-[#163d8a]
            transition text-[18px]">Statics Modules</Link>
            <p className="text-left text-black text-[18px]
            font-bold">Chapter 2: Force Systems</p>
            <Link href="/2D-solver" className="bg-[#1848a0] text-white
            px-6 py-3 rounded-md shadow hover:bg-[#163d8a] transition
            text-[18px]">Resultant Force Calculator</Link>
            <p className="text-left text-black text-[18px]
            font-bold">Chapter 3: Equilibrium</p>
            <Link href="/Equilibrium" className="bg-[#1848a0] text-white
            px-6 py-3 rounded-md shadow hover:bg-[#163d8a] transition
            text-[18px]">Equilibrium Calculator</Link>
            <p className="text-left text-black text-[18px]
            font-bold">Chapter 4: Structures</p>
            <Link href="/Structures" className="bg-[#1848a0] text-white
            px-6 py-3 rounded-md shadow hover:bg-[#163d8a] transition
            text-[18px]">Truss Calculator</Link>
            <p className="text-left text-black text-[18px]
            font-bold">Chapter 5: Distributed Loads</p>
            <Link href="/Distributed-Loads" className="bg-[#1848a0]
            text-white px-6 py-3 rounded-md shadow hover:bg-[#163d8a]
            transition text-[18px]">Structures Calculator</Link>
          </div>
```

```
        </div>
      </section>
      <Footer/>
    </main>
  );
}
```

Introduction

```
import Link from "next/link";
import Header from "../components/Header";
import Footer from "../components/Footer";

export default function HomePage(){
  return(
    <main className="bg-gray-50 min-h-screen flex flex-col">
      <Header/>
      <section className="flex flex-col items-center text-center py-12
      bg-gray-50 flex-grow">
        <h1 className="text-[20px] md:text-[80px] font-bold mb-2
        text-black">
          Stati<span className="text-[#008409]">Calcs</span>
        </h1>
        <p className="text-gray-600 mb-12 text-[18px]">
          Modules for learning and solving Statics of Rigid Bodies.
        </p>
        <div className="w-full max-w-2xl">
          <div className="grid grid-cols-2 gap-4 items-center">
            <p className="text-left text-black text-[18px]
            font-bold">Module 1: Free Body Diagram</p>
            <Link href="/modules/FBD" className="bg-[#008409] text-white
            px-6 py-3 rounded-md shadow hover:bg-[#15711b] transition
            text-[18px]">Module 1</Link>
            <p className="text-left text-black text-[18px]
            font-bold">Module 2: 2D Resultant</p>
            <Link href="/modules/2D-Resultant" className="bg-[#008409]
            text-white px-6 py-3 rounded-md shadow hover:bg-[#15711b]
            transition text-[18px]">Module 2</Link>
            <p className="text-left text-black text-[18px]
            font-bold">Module 3: 3D Resultant</p>
            <Link href="/modules/3D-Resultant" className="bg-[#008409]
            text-white px-6 py-3 rounded-md shadow hover:bg-[#15711b]
            transition text-[18px]">Module 3</Link>
            <p className="text-left text-black text-[18px]
            font-bold">Module 4: 2D Equilibrium</p>
            <Link href="/modules/2D-Equilibrium" className="bg-[#008409]
            text-white px-6 py-3 rounded-md shadow hover:bg-[#15711b]
            transition text-[18px]">Module 4</Link>
```

```
            <p className="text-left text-black text-[18px]
            font-bold">Module 5: 3D Equilibrium</p>
            <Link href="/modules/3D-Equilibrium" className="bg-[#008409]
            text-white px-6 py-3 rounded-md shadow hover:bg-[#15711b]
            transition text-[18px]">Module 5</Link>
            <p className="text-left text-black text-[18px]
            font-bold">Module 6: Truss Joint Method</p>
            <Link href="/modules/Truss" className="bg-[#008409]
            text-white px-6 py-3 rounded-md shadow hover:bg-[#15711b]
            transition text-[18px]">Module 6</Link>
            <p className="text-left text-black text-[18px]
            font-bold">Module 7: Moment of Inertia about Centroidal
            Axes</p>
            <Link href="/modules/MOIcentroidal" className="bg-[#008409]
            text-white px-6 py-3 rounded-md shadow hover:bg-[#15711b]
            transition text-[18px]">Module 7</Link>
            <p className="text-left text-black text-[18px]
            font-bold">Module 8: Moment of Inertia about Custom Axes</p>
            <Link href="/modules/MOIcustom" className="bg-[#008409]
            text-white px-6 py-3 rounded-md shadow hover:bg-[#15711b]
            transition text-[18px]">Module 8</Link>
          </div>
        </div>
      </section>
      <Footer/>
    </main>
  );
}
```

2D Resultant Force Calculator

```
"use client";
import { useRef, useState } from "react";
import Header from "../../components/Header";
import Footer from "../../components/Footer";
import "katex/dist/katex.min.css";
import { useStepByStepPDF } from "../ToPDF/Page";
import { StepByStepSolution, fromLegacySteps } from
"../../components/StepByStep";

const fmt2 = (v: number): string => {
  if (Math.abs(v - Math.round(v)) < 1e-9) return
  Math.round(v).toString();
  return v.toFixed(2).replace(/\.?0+$/, "");
};

class ForceSystem2D {
  vectors: { fx: number; fy: number; magnitude: number; angleDeg: number
  }[];
```

```
  constructor() { this.vectors = []; }
  addForce(magnitude: number, angleDeg: number) {
    const angleRad = angleDeg * Math.PI / 180;
    const fx = magnitude * Math.cos(angleRad);
    const fy = magnitude * Math.sin(angleRad);
    this.vectors.push({ fx, fy, magnitude, angleDeg });
  }
  stepByStepSolution() {
    const steps: string[] = [];
    let sumFx = 0, sumFy = 0;
    steps.push("Step 1: Resolve each force into components:");
    this.vectors.forEach((v, i) => {
      steps.push(`F_{x${i+1}} = ${v.magnitude}cos(${v.angleDeg}) =
      ${fmt2(v.fx)} kN`);
      steps.push(`F_{y${i+1}} = ${v.magnitude}sin(${v.angleDeg}) =
      ${fmt2(v.fy)} kN`);
      sumFx += v.fx; sumFy += v.fy;
    });
    const R = Math.hypot(sumFx, sumFy);
    const theta = Math.atan2(sumFy, sumFx) * 180 / Math.PI;
    steps.push("Step 2: Sum of components:");
    steps.push(`SigmaFx = ${fmt2(sumFx)} kN`);
    steps.push(`SigmaFy = ${fmt2(sumFy)} kN`);
    steps.push("Step 3: Resultant force:");
    steps.push(`R = ${R.toFixed(3)} kN`);
    steps.push(`theta = ${theta.toFixed(2)} degrees`);
    return { steps, sumFx, sumFy, R, theta };
  }
}

export default function Solver2D() {
  const [forces, setForces] = useState([{ magnitude: "", angle: "" }]);
  const [result, setResult] = useState(null);
  const calculateResultant = () => {
    const system = new ForceSystem2D();
    forces.forEach((f) => {
      const m = parseFloat(f.magnitude);
      const a = parseFloat(f.angle);
      if (!isNaN(m) && !isNaN(a)) system.addForce(m, a);
    });
    setResult(system.stepByStepSolution());
  };
  return (
    <div><Header/>
      <main>
        <button onClick={calculateResultant}>Calculate</button>
      </main>
    <Footer/></div>
  );
}
```

3D Resultant Force Calculator – Angles Tab

```
"use client";
import { useRef, useEffect, useState } from "react";
import * as THREE from "three";

class ForceSystem3D {
  vectors: any[] = [];
  addForce(magnitude: number, azimuthDeg: number, elevationDeg: number)
  {
    const az = azimuthDeg * Math.PI / 180;
    const el = elevationDeg * Math.PI / 180;
    this.vectors.push({
      fx: magnitude * Math.cos(el) * Math.cos(az),
      fy: magnitude * Math.cos(el) * Math.sin(az),
      fz: magnitude * Math.sin(el),
      magnitude, azimuthDeg, elevationDeg
    });
  }
  solve() {
    let sumFx = 0, sumFy = 0, sumFz = 0;
    this.vectors.forEach((v) => {
      sumFx += v.fx; sumFy += v.fy; sumFz += v.fz;
    });
    const R = Math.sqrt(sumFx**2 + sumFy**2 + sumFz**2);
    const azimuth = Math.atan2(sumFy, sumFx) * 180 / Math.PI;
    const elevation = Math.asin(sumFz / (R || 1)) * 180 / Math.PI;
    const alpha = Math.acos(sumFx / R) * 180 / Math.PI;
    const beta = Math.acos(sumFy / R) * 180 / Math.PI;
    const gamma = Math.acos(sumFz / R) * 180 / Math.PI;
    return { sumFx, sumFy, sumFz, R, azimuth, elevation, alpha, beta,
    gamma };
  }
}

export default function AnglesTab() {
  const [forces, setForces] = useState([{ magnitude: "", azimuth: "",
  elevation: "" }]);
  const [result, setResult] = useState(null);
  const calculate = () => {
    const sys = new ForceSystem3D();
    forces.forEach(f => {
      const m = parseFloat(f.magnitude), az = parseFloat(f.azimuth), el
      = parseFloat(f.elevation);
      if (!isNaN(m) && !isNaN(az) && !isNaN(el)) sys.addForce(m, az,
      el);
    });
    setResult(sys.solve());
  };
  return (<div><button onClick={calculate}>Calculate</button></div>);
}
```

3D Resultant Force Calculator – Coordinates Tab

```
"use client";
import { useRef, useEffect, useState } from "react";
import * as THREE from "three";

function buildSolution(details: any[], Rx: number, Ry: number, Rz:
number, R: number) {
  const steps = [];
  steps.push("Step 1: Determine position vectors");
  details.forEach(d => steps.push(
    `r_${d.from}${d.to} = (${d.bx}-${d.ax})i + (${d.by}-${d.ay})j +
    (${d.bz}-${d.az})k`
  ));
  steps.push("Step 2: Unit vectors");
  details.forEach(d => steps.push(
    `u_${d.from}${d.to} = r_${d.from}${d.to} / ${d.len.toFixed(2)}`
  ));
  steps.push("Step 3: Force vectors");
  details.forEach(d => steps.push(
    `F_${d.to} = ${d.mag} * u_${d.from}${d.to} = ${d.Fx.toFixed(2)}i +
    ${d.Fy.toFixed(2)}j + ${d.Fz.toFixed(2)}k`
  ));
  steps.push("Step 4: Resultant");
  steps.push(`R = ${Rx.toFixed(2)}i + ${Ry.toFixed(2)}j +
  ${Rz.toFixed(2)}k`);
  steps.push(`|R| = ${R.toFixed(2)} kN`);
  return steps;
}

export default function CoordinateTab() {
  const [points, setPoints] = useState([
    { label: "A", x: "", y: "", z: "" },
    { label: "B", x: "", y: "", z: "" }
  ]);
  const [forces, setForces] = useState([{ mag: "", from: 0, to: 1 }]);
  const [result, setResult] = useState(null);
  const calculate = () => {
    let Rx = 0, Ry = 0, Rz = 0;
    const details = [];
    forces.forEach(f => {
      const mag = parseFloat(f.mag);
      if (!mag) return;
      const a = points[f.from], b = points[f.to];
      if (!a || !b) return;
      const dx = (parseFloat(b.x) || 0) - (parseFloat(a.x) || 0);
      const dy = (parseFloat(b.y) || 0) - (parseFloat(a.y) || 0);
      const dz = (parseFloat(b.z) || 0) - (parseFloat(a.z) || 0);
      const len = Math.sqrt(dx*dx + dy*dy + dz*dz);
      if (!len) return;
      const Fx = mag*dx/len, Fy = mag*dy/len, Fz = mag*dz/len;
      Rx += Fx; Ry += Fy; Rz += Fz;
```

```
      details.push({ mag, from: a.label, to: b.label, dx, dy, dz, Fx, Fy,
      Fz, len });
    });
    const R = Math.sqrt(Rx*Rx + Ry*Ry + Rz*Rz);
    setResult({ steps: buildSolution(details, Rx, Ry, Rz, R), Rx, Ry, Rz,
    R });
  };
  return (<div><button onClick={calculate}>Calculate</button></div>);
}
```

Equilibrium of Concurrent Force System

```
"use client";
import { useRef, useState, useEffect } from "react";
import Header from "../../components/Header";
import Footer from "../../components/Footer";
import { solveEquilibrium2D } from "../../lib/Equilibriumconc";

type ForceInput = {
  magnitude: string;
  angle: string;
  magnitudeUnknown?: boolean;
  angleUnknown?: boolean;
};

function calculate(forces: ForceInput[]) {
  const magUnknownIndices = forces.map((f, i) => f.magnitudeUnknown ? i :
 -1).filter(i => i !== -1);
  const angUnknownIndices = forces.map((f, i) => f.angleUnknown ? i :
 -1).filter(i => i !== -1);
  const totalUnknowns = magUnknownIndices.length +
  angUnknownIndices.length;
  if (totalUnknowns > 2) return { error: "Too many unknowns" };
  const knownForces = forces
    .filter((f, i) => !magUnknownIndices.includes(i) &&
    !angUnknownIndices.includes(i))
    .map((f, i) => {
      const magnitude = parseFloat(f.magnitude);
      const angle = parseFloat(f.angle);
      if (isNaN(magnitude) || isNaN(angle)) return null;
      return { magnitude, angle, label: `F_${i+1}` };
    }).filter(Boolean);
  return { knownForces };
}

export default function Equilibrium() {
  const [forces, setForces] = useState([{ magnitude: "", angle: "" }]);
  const [solution, setSolution] = useState(null);
  const [error, setError] = useState(null);
```

```
  const handleCalculate = () => {
    const result = calculate(forces);
    if ("error" in result) { setError(result.error); return; }
    setSolution(result);
  };
  return (
    <div><Header/>
      <main>
        <button onClick={handleCalculate}>Calculate</button>
      </main>
    <Footer/></div>
  );
}
```

Equilibrium of Non-Concurrent Force System – Beam Solver

```
"use client";
import { useState } from "react";
import { solveBeam, BeamResult } from "../../lib/beamEngine";

type Support = { type: "Pinned" | "Roller"; location: string };
type PointLoad = { magnitude: string; location: string };
type DistributedLoad = { start: string; end: string; startMag: string;
endMag: string };

export default function BeamSolverUI() {
  const [beamLength, setBeamLength] = useState("");
  const [supports, setSupports] = useState([{ type: "Pinned", location:
  "" }]);
  const [pointLoads, setPointLoads] = useState([{ magnitude: "",
  location: "" }]);
  const [distributedLoads, setDistributedLoads] = useState([
    { start: "", end: "", startMag: "", endMag: "" }
  ]);
  const [result, setResult] = useState<BeamResult | null>(null);
  const [error, setError] = useState<string | null>(null);
  const calculate = () => {
    setError(null);
    try {
      const res = solveBeam({
        beamLength: parseFloat(beamLength),
        supports: supports.map(s => ({ type: s.type, location:
        parseFloat(s.location) })),
        pointLoads: pointLoads.map(p => ({
          magnitude: parseFloat(p.magnitude),
          location: parseFloat(p.location)
        })),
        distributedLoads: distributedLoads.map(d => ({
          start: parseFloat(d.start),
```

```
          end: parseFloat(d.end),
          startMag: parseFloat(d.startMag),
          endMag: parseFloat(d.endMag)
        }))
      });
      setResult(res);
    } catch (e: any) { setError(e.message); }
  };
  return (
    <div>
      <h1>Beam Solver</h1>
      <button onClick={calculate}>Calculate</button>
    </div>
  );
}
```

Truss Solver

```
"use client";
import { useState } from "react";
import Header from "../../components/Header";
import Footer from "../../components/Footer";
import { solveTruss, type Support, type Joint, type Member, type Force,
type Solution } from "../../lib/truss";

export default function TrussSolverUI() {
  const [supports, setSupports] = useState([
    { x: "", y: "", type: "Pinned" },
    { x: "", y: "", type: "Roller" }
  ]);
  const [nodes, setNodes] = useState([{ x: "", y: "" }]);
  const [members, setMembers] = useState([{ start: 0, end: 1 }]);
  const [forces, setForces] = useState([{ Joint: 0, magnitude: "", angle:
  "" }]);
  const [solution, setSolution] = useState<Solution | null>(null);
  const handleSolve = () => {
    setSolution(solveTruss(supports, nodes, members, forces));
  };
  return (
    <div><Header/>
      <main>
        <button onClick={handleSolve}>Calculate</button>
      </main>
    <Footer/></div>
  );
}
```

Moment of Inertia Calculator

```
"use client";
import { useState } from "react";
import Header from "../../components/Header";
import Footer from "../../components/Footer";
import ShapeCanvas from "../../components/ShapeCanvas";
import { computeMOI } from "../../lib/MOIEngine";
import { computeCustomAxis } from "../../lib/MOIcustom";
import type { ShapeType } from "../../types/shapes";

type XY = { x: string; y: string };
type ShapeData = {
  type: ShapeType;
  hollow: "Hollow" | "Solid";
  isOpen: boolean;
  nodes: XY[];
  sides: { a: number; b: number }[];
  radius: string;
  x: string;
  y: string;
};

export default function DistributedLoadPage() {
  const [axisType, setAxisType] = useState("Centroidal");
  const [axisX, setAxisX] = useState("");
  const [axisY, setAxisY] = useState("");
  const [shapes, setShapes] = useState([{
    type: "Polygon", hollow: "Solid", isOpen: true,
    nodes: [{ x: "", y: "" }, { x: "", y: "" }],
    sides: [{ a: 0, b: 1 }], radius: "", x: "", y: ""
  }]);
  const [result, setResult] = useState(null);
  const calculateResultant = () => {
    const centroidal = computeMOI(shapes);
    let customMOI;
    if (axisType === "Custom") {
      customMOI = computeCustomAxis([], Number(axisX), Number(axisY));
    }
    setResult({
      centroid: centroidal.centroid,
      centroidMOI: centroidal.final,
      customMOI
    });
  };
  return (
    <div><Header/>
      <main>
        <ShapeCanvas shapes={shapes} axisType={axisType}
        axisX={Number(axisX)} axisY={Number(axisY)}/>
        <button onClick={calculateResultant}>Calculate</button>
      </main>
```

```
    <Footer/></div>
  );
}
```

Module Page

```
"use client";
import Header from "../components/Header";
import Footer from "../components/Footer";

function ArrowItem({ label }: { label: string }) {
  return (
    <li className="flex items-center gap-2">
      <span className="text-[#1848a0] font-bold">arrow</span>{label}
    </li>
  );
}

function Step({ number, title, children }: { number: number; title:
string; children: React.ReactNode }) {
  return (
    <div className="mb-6">
      <div className="flex items-center gap-3 mb-3">
        <div className="flex-shrink-0 w-9 h-9 rounded-full bg-[#1848a0]
        text-white flex items-center justify-center font-bold
        text-[16px]">
          {number}
        </div>
        <h2 className="text-[20px] font-semibold">{title}</h2>
      </div>
      <div className="ml-12 bg-white border border-gray-200 rounded-2xl
      shadow p-5 text-[18px] space-y-3">
        {children}
      </div>
    </div>
  );
}

export default function Equilibrium2D() {
  return (
    <div className="relative flex flex-col min-h-screen bg-gray-50
    text-gray-900 text-[18px]">
      <Header/>
      <main className="flex-grow flex flex-col items-center px-4 py-10">
        <h1 className="text-[32px] font-bold mb-2 text-center">2D
        Equilibrium Problems</h1>
        <p className="text-gray-500 mb-8 text-center">Procedure for
        Solving 2D Equilibrium Problems</p>
        <div className="w-full max-w-xl">
```

```
        <Step number={1} title="Draw the Free Body Diagram (FBD)">
          <p>Isolate the body and represent all external forces and
          moments.</p>
        </Step>
        <Step number={2} title="Choose Coordinate System">
          <p>Define x- and y-axes.</p>
        </Step>
        <Step number={3} title="Resolve Forces into Components">
          <p>Decompose forces: Fx = F cos theta, Fy = F sin theta</p>
        </Step>
        <Step number={4} title="Apply Equilibrium Equations">
          <p>SigmaFx = 0, SigmaFy = 0, SigmaM = 0</p>
        </Step>
        <Step number={5} title="Solve for Unknowns">
          <p>Use substitution or elimination.</p>
        </Step>
        <Step number={6} title="Check Results">
          <p>Verify units, signs, and equations.</p>
        </Step>
      </div>
    </main>
    <Footer/>
  </div>
 );
}
```

Curriculum Vitae

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Educational Background

Primary

New Mabuhay Elementary School
55G4+W24, General Santos City (Dadiangas), South Cotabato
2009-2013

Upper Tambler Central Elementary School
Purok 6 Teacher's Village, General Santos City (Dadiangas), 9500 South Cotabato
2013-2014

Habitat Community Elementary School
Habitat Phase A, General Santos City (Dadiangas), 9500 South Cotabato
2014

Upper Tambler Central Elementary School
Purok 6 Teacher's Village, General Santos City (Dadiangas), 9500 South Cotabato
2014-2026

Secondary

Junior High School
Lagao National High School
45MC+V66, Aparente St, General Santos City (Dadiangas), 9500 South Cotabato
2016-2020

Senior High School
Sped Integrated School
Purok Malipayon Brgy, General Santos City (Dadiangas), 9500 South Cotabato
2020-2022

Tertiary

MINDANAO STATE UNIVERSITY GENERAL SANTOS CITY
Bachelor of Science in Civil Engineering
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Curriculum Vitae

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Educational Background

Primary	**UCCP Christian School Polomolok, Inc.** Corner Frech Penido St., Polomolok, South Cotabato 2010-2016
Secondary	*Junior High School* **Poblacion Polomolok National High School** Barangay Pagalungan, Polomolok, South Cotabato 2016-2020
	Senior High School **Mindanao State University – Gensan Senior High School** Brgy. Fatima, GSC 2020-2022
Tertiary	**MINDANAO STATE UNIVERSITY GENERAL SANTOS CITY** Bachelor of Science in Civil Engineering Brgy. Fatima, GSC 2022 - present

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Scholarships

- PAeskwela TAbang sa BAtang PObre (PATABA-PO) Scholarship

Honors and Awards

- Prettiest Civil Engineering Student
  MSU-GSC, May 2026